

SEMESTER 3

PRODUCTION ENGINEERING

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hr. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9

3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

SOLID MECHANICS AND DESIGN

Course Code	PCPET302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

- 1.To enable students to grasp the essential concepts of stress, strain, and their interrelationships, while identifying and describing the mechanical properties of engineering materials.
- 2.To familiarize the principles of stress, strain and deformation to analyze and solve problems in mechanical components and structures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concept of stress, Types of Stress, Uniaxial Tension Test, Stress-Strain Diagram, Ductile & Brittle Materials, Conditions affecting Mechanical Properties, Hardness and Impact Strength. Cartesian and Cylindrical Coordinate Systems, Stress at a point, Complementary shear stress, concept of strain, shear strain, generalized Hook's Law. Principal strain in terms of stress, plain stress, hydrostatic stress and plane strain, stress-strain relations; linear elastic, rigid, perfectly plastic (non-strain hardening), rigid plastic (strain hardening), elastic perfectly plastic, elastic-plastic, isotropy, elastic constants and relationship among them, overview of impact test and Hardness Test; Rockwell, Brinel &Vickers hardness test. Concepts of creep and fatigue.	11

2	Members subjected to uniaxial stress- bars subjected to tension and compression, compound bars subjected to temperature change, strain energy in uniaxial loading, Moduli of resilience and toughness. Shear Force and Bending Moment Diagrams- Sign convention, Relationship Between Load, Shear Force and Bending Moment.	11
3	Flexural and shear stresses in beams- members subjected to flexural loads, pure bending, derivation of elastic flexural formula, shear stress in beams of typical cross sections. Principal stresses and strains-stress on oblique planes, material subjected to stress in two mutually perpendicular planes, materials subjected combined direct and shear stress, Mohr's Circle.	11
4	Factor of safety, Failure Theories (overview). Torsion theory, torsion of solid and hollow circular shafts, distribution of shear stress in shafts. power transmitted by a solid stepped and hollow shaft, design of shafts subjected to static loading (design for strength and rigidity). ASME code, Design of sunk keys, Introduction to design of Rigid and flexible coupling.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Distinguish the response of materials under the action of various types of stresses	K3
CO2	Model stress distribution in loaded beams of various types	K3
CO3	Define stress system and assess response of loaded members with respect to geometry	K3
CO4	Design of basic power transmission elements	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	-	-	3
CO2	3	3	3	2	-	-	-	-	-	-	-	1
CO3	3	3	3	1	-	-	-	-	-	-	-	1
CO4	3	3	3	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials	S. S. Rattan	McGraw Hill publishers	3rd Ed. 2017
2	Design of Machine elements	V. B. Bhandari	McGraw Hill publishers	5th Ed. 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of Materials	Ferdinand P. Beer	McGraw Hill	8th Ed.2020
2	Mechanics of Materials	Egor P. Popov	Pearson	2nd Ed. 2015
3	Strength of Materials	S. S. Bhavikatti	Vikas publishing House	5th Ed.2021
4	Shigley's Mechanical Engineering Design	Richard G. Budynas, J. Keith Nisbett	McGraw Hill	11th Ed. 2020

Data book			
Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
Design Data Handbook- For Mechanical Engineers	K. Mahadevan, K. Balaveera Reddy	CBS Printers and distributors	4th Ed. 2019
Design Data-Data book for Engineers	Faculty of Mechanical Engineering -PSG College of Technology	Kalaikathir Achchagam, Coimbatore	5th Ed. 2023

Video Links	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107146/
2	Lectures on the Chapters of the textbook "Mechanics of Materials", Ferdinand P. Beer, E. Russell Johnston Jr., John T. DeWolf & David F. Mazurek. https://www.youtube.com/watch?v=GuIYMVHvnhs&list=PLAjJ9epAHCKz-vpeUJatstaKOtBHHyZI_&pp=iAQB
3	Machine Design -Prof. Ramesh Singh(IIT Bombay) Web courseware https://www.me.iitb.ac.in/~ramesh/courses/ME423/me423.html
4	Design of Machine elements(NPTEL Tutorials) https://youtube.com/playlist?list=PLfdm9m79os6LxsLyBxQP2PjF-brqYkg_F&feature=shared

SEMESTER S3

FLUID AND THERMAL SCIENCES

Course Code	PCPET303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the fundamental concepts and methods of fluid mechanics and thermal science
2. To equip students to apply fluid mechanics and thermal science concepts for practical applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Physical properties of fluids, density, specific weight, vapour pressure, Newton's law of viscosity. Newtonian and non-Newtonian fluids. Fluid Statics: Fluid pressure, Forces on solid surfaces, Buoyant forces. Kinematics of Fluid Flow: Equations for acceleration, Continuity equation, Irrotational and rotational flow, Potential and stream functions.	11
2	Dynamics of Fluid Flow: Euler and Bernoulli's theorems, Impulse momentum theory, Applications of energy and momentum equations. Pipe Flow Systems: Darcy-Weisbach equation, Energy losses in pipelines, Minor losses. Thermodynamics: Thermodynamic System and Control Volume, Surrounding, Boundaries, Thermodynamic properties, Process, Cycle, Thermodynamic Equilibrium, Quasi-static Process, State, Point and Path function. Zeroth Law of Thermodynamics.	11

3	Energy and Work: PdV work and other types of work transfer, free expansion work, heat and heat capacity. First law of Thermodynamics and applications - Enthalpy- specific heats. Second Law of Thermodynamics, Thermal Reservoir, Heat Engine, Heat pump – Kelvin- Planck and Clausius Statements, Reversibility, Irreversible Process, Carnot's theorem and its corollaries, Absolute Thermodynamic Temperature scale. Clausius Inequality, Entropy- Entropy changes in various thermodynamic processes, principle of increase of entropy and its applications, Entropy generation, Third law of thermodynamics.	11
4	Overview of heat transfer mechanisms: Introductory concepts of conduction, convection, radiation. Fourier's law of heat conduction. Newton's law of cooling. Conduction: Steady state heat conduction in plane walls, Multi layer walls. Heat conduction in cylinders. Thermal resistance concept. Critical radius of insulation. Heat transfer from fin surfaces. Fin efficiency, Fin effectiveness. Convection: Free and forced convection. Physical mechanism of forced convection, convection heat transfer coefficient, Dimensionless numbers- Nusselt number, Velocity boundary layer, Laminar and turbulent flows, Reynolds number, Thermal boundary layer, Prandtl number. Natural convection, Grashof number. Thermal radiation: black body radiation – radiation properties – laws of thermal radiation, Stefan – Boltzmann law, Wien's displacement law, black and grey surfaces. View factor.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Demonstrate the fundamental concepts of fluid mechanics, including pressure, hydrostatics, and buoyancy	K2
CO2	Solve fluid systems and design problems; Explain basic thermodynamics concepts	K3
CO3	Apply the principles of heat transfer, thermodynamics, and energy conversion	K3
CO4	Make use of thermal systems and design solutions	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	-	-	-	2
CO2	3	3	3	-	-	-	-	-	-	-	-	2
CO3	3	3	3	-	-	-	-	-	-	-	-	2
CO4	3	3	3	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Thermodynamics	P. K Nag	McGraw Hill Education	6th Ed. 2017
2	Heat and Mass Transfer	P. K Nag	McGraw Hill Education	3rd Ed. 2011
3	Fluid Mechanics	R. K. Bansal	Laxmi Publications	2ndEd. 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Thermodynamics: An Engineering Approach	Yunus A Cengel	Michael A Boles	9th Ed. 2019
2	Heat Transfer	Holman, J.P.	Tata Mc Graw Hill	9th Ed. 2017
3	Fluid Mechanics	R.C Hibbeler	Pearson education	2ndEd. 2022

SEMESTER S3

MATERIAL SCIENCE AND ENGINEERING

Course Code	PBPET304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	4	ESE Marks	40
Credits	3:0:0:1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize about the structure, and properties, of various engineering materials.
2. To enable students for identification of appropriate material for a given engineering application.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to materials- Classification- Crystal Structures -SC, BCC, FCC, HCP structures, APF - theoretical density simple problems – Miller Indices: - crystal plane and direction - Modes of plastic deformation: - Slip and twinning -Schmid's law - Crystal Imperfections, Point Defects- Line Defects-Surface Defects-Bulk Defects. Critical nucleus size and Critical Free energy- Nucleation-Homogeneous and Heterogeneous Nucleation- Growth - Single crystal -Polycrystalline Materials - Basic principles of solidification of metals and alloys. Growth of crystals- Planar growth – dendritic growth – Solidification time - Cooling curves - non-crystalline solids- Glass Transition Temperature. Effects of grain size, Hall - Petch theory, simple problems.	11
2	Diffusion in solids, Fick's laws, mechanisms, applications of diffusion in mechanical engineering, simple problems. Solutions-Hume Rothery Rules, Phase Diagrams: Phase Rule –Unary System- Binary Phase diagrams- Isomorphous systems- Congruent phase diagrams - Tie Line- Lever Rule- Eutectic, Peritectic, Eutectoid and Peritectoid reactions.	11

	Typical Phase diagrams –Fe-Fe ₃ C system- Cu-Zn System – Pb-Sn system- Ag-Pt System-Iron-Iron carbide Equilibrium Diagram-perform phase equilibrium calculation and construction of phase diagram.	
3	Recovery recrystallization and grain growth, cold and hot working- Bauschinger effect. Heat treatment: - Definition and necessity – TTT for a eutectoid iron–carbon alloy, CCT diagram, applications - annealing, normalizing, hardening, spheroidizing. Tempering: - Austempering, martempering and Aus forming. Hardenability, Jominy end quench test, applications- Surface hardening methods. Alloy steels: effect of alloying elements on properties of steel, strengthening mechanisms. Characteristics of fatigue failure, fatigue tests, S-N curve. Fracture: – Brittle and ductile fracture – Griffith theory of brittle fracture – Stress concentration, stress raiser – Effect of plastic deformation on crack propagation – trans-granular, inter-granular fracture. Ductile to brittle transition temperature (DBTT).	11
4	Introduction to composite materials along with its basic requirements and classification; metal matrix (MM), polymer Matrix (PM) and ceramic matrix (CM) composites, Secondary processing and joining of various composite materials. Introduction to Nanomaterials: Classification of nanostructures, nanoscale architecture, fundamental structure and their properties. Smart Materials: Piezoelectric Materials, Electrostrictive Materials, Magnetostrictive Materials, Magnetoelectric Materials. Magnetorheological Fluids, Electrorheological Fluids, Shape Memory Materials, Optic Fibers. Applications of smart materials.	11

Suggestion on Project Topics

- Selection of biodegradable materials for packing industries.
- Investigation of the mechanical properties of graphene composites.
- Development of coatings for corrosion resistance on steel surfaces.
- Fabrication and testing of shape memory alloys for applications in actuators.
- Analysis of the mechanical performance of natural fiber-reinforced composites.
- Design and testing of hydrogels that respond to environmental stimuli.

- Evaluation of the mechanical properties of 3D printed materials versus traditional manufacturing methods.
- Development of novel materials for battery applications, such as lithium-ion batteries.
- Investigation of surface treatments to improve wear resistance in materials.
- Analysis of the feasibility of recycling metal waste in manufacturing processes.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Analyze crystal structures for identifying the crystal defects, and differentiate homogeneous and heterogeneous nucleation.	K4
CO2	Explain Hume Rothery rules and understand the equilibrium diagram and invariant reactions.	K2
CO3	Explain TTT diagram, heat treatment processes and differentiate cold and hot working.	K2
CO4	Identify advanced materials and their applications.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	2	3	-	-	-	-	-	-	3
CO2	3	2	-	2	2	-	-	-	-	-	-	3
CO3	3	3	-	3	2	-	-	-	-	-	-	3
CO4	3	3	-	3	3	-	-	-	-	-	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Science and Engineering	Callister William. D	John Wiley,	10th Ed. 2014
2	Elements of Material Science	Van Vlack	Addison Wesley	6th Ed. 1989

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Science for Engineers	J.C. Anderson, Keith D. Leaver, Rees D. Rawlings, Patrick S. Leever	CRC Press	5th Ed. 2005
2	Fundamentals of Materials Science and Engineering: An Integrated Approach	Callister, William D. Jr.	John Wiley and Sons	2nd Ed. 2003
3	Material Science and Engineering	Raghavan V	Prentice Hall	5th Ed. 2004
4	Nanoscale Science and Technology	Robert K, Ian H, Mark G	John Wiley & sons Ltd	1st Ed. 2005
5	Heat Treatment Principles and Techniques	Rajan T. V	PHI Learning Pvt. Ltd.	3rd Ed. 2010

Video Links	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/113/102/113102080/
2	https://archive.nptel.ac.in/courses/113/102/113102080/
3	https://archive.nptel.ac.in/courses/113/102/113102080/
4	https://archive.nptel.ac.in/courses/112/104/112104229/ https://archive.nptel.ac.in/courses/113/104/113104102/ https://archive.nptel.ac.in/courses/112/104/112104251/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value	11

	Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	
3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	AurélienGéron	O'Reilly Media, Inc.	2 nd edition,2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmn nnibpcajpcglclefindm kaj/https://www.math. arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP_{fc} from GDP_{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP_{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, $NFIA=(-200)$ and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of confidentiality in moral integrity, Codes of Ethics .	6

	Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and	6

	climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	6

**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.

- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group /Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.

- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India

highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

COMPUTER AIDED DESIGN LAB

Course Code	PCPEL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarize the basic concepts to drawing, edit, dimension, hatching etc. to develop 2D & 3D Modelling.
2. To enable students to create 3D modelling, Assembling, modification & manipulation along with detailing.

Expt. No.	Experiments (Minimum of 10 experiments should be completed)
1	Simple 2D models using drawing and editing tools. (foundation bolts)
2	Advanced 2D models using drawing and editing tools. (cotter joint with bill of materials)
3	Advanced 2D models using drawing and editing tools. (knuckle joint with bill of materials)
4	Advanced 2D models using drawing and editing tools. (flanged coupling with bill of materials)
5	Advanced 2D models using drawing and editing tools. (flexible coupling with bill of materials)
6	Isometric drawings using 2D modelling.
7	Simple Engineering parts using 3D modelling software
8	3D assembly (Plummer block)
9	3D assembly (Connecting rod)
10	3D assembly (Machine vice)
11	3D assembly (Screw jack)
12	3D assembly (Lathe tailstock)

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Develop 2D and 3D models using basic tools in CAD software .	K2
CO2	Familiarize with various commands in the tools box to develop 2D & 3D models.	K3
CO3	Build, modify and manipulate 3D assemblies of machine components .	K3
CO4	Develop 2D drafting from 3D part models and their assemblies.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3	-	-	-	3	3	-	3
CO2	3	2	2	2	3	-	-	-	3	3	-	3
CO3	3	2	2	2	3	-	-	-	3	3	-	3
CO4	3	2	2	2	3	-	-	-	3	3	-	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Machine Drawing	P.I. Varghese & K.C. John	VIP Publishers	2015
2	AutoCAD 2017	Prof. Sham Tickoo	Cad cim Technologies	23rd Ed. 2016
3	Creo Parametric 2.0 for Designers	Prof. Sham Tickoo	Cad cim Technologies	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Machine Drawing	Bhat N. D., Panchal	Charotar Pub. House	50th Ed. 2000.
2	Machine Drawing	Narayana K. L., Kannaiah P.	New age international Publishers	5th Ed. 2000
3	A Textbook of Machine Drawing	Gill P. S.,	K. Kataria and Sons, New Delhi	18th Ed. 2008

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

CONTINUUM MECHANICS LAB

Course Code	PCPEL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarize the physical properties of fluids, including density, viscosity, and surface tension, and their impact on fluid flow in different scenarios with hands-on training.
2. To familiarize the mechanical properties of various materials, such as tensile strength, compressive strength, hardness, and toughness, through standard testing methods.

Expt. No.	Experiments (Minimum 10 experiments should be completed)
1	Tension test on Mild Steel using Universal Testing Machine and Extensometers.
2	Tests on springs (Open and Close coiled).
3	Hardness test (Brinell, Vickers and Rockwell).
4	Impact test (Izod and Charpy).
5	Torsion test on Mild Steel rods.
6	Strut test (Column buckling experiment).
7	Determination of moment of inertia of rotating bodies.
8	Fatigue life (endurance) test.
9	Verification of Bernoulli's theorem.
10	Calibration of Venturi meter.
11	Calibration of Orifice meter.
12	Estimation of hydraulic coefficients of orifices and mouth pieces.
13	Calibration of triangular and rectangular notches.
14	Determination of loss of head in a pipeline.
15	Determination of friction factor for a given pipe line.
16	Determination of Metacentric height.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva Voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Determine the strength of materials upon normal external loads.	K3
CO2	Identify the mechanical behaviour of materials under different loading conditions	K3
CO3	Evaluate the performance of different instruments and measurement devices.	K3
CO4	Determine the hydraulic parameters of different flow measurement devices.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	1	-	1	-	-	2	2	-	2
CO2	3	3	3	2	-	1	-	-	2	2	-	2
CO3	3	3	3	-	-	1	-	-	2	2	-	2
CO4	3	3	3	-	-	1	-	-	2	2	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials	S. S. Rattan	McGraw Hill publishers	3rd Ed. 2017
2	Mechanics of Materials	Ferdinand P. Beer,	McGraw Hill	8th Ed. 2020
3	Fluid Mechanics	R.K. Bansal	Laxmi Publications	2nd Ed. 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of Materials	Egor P. Popov	Pearson	2nd Ed. 2015
2	Fluid Mechanics	R.C Hibbeler	Pearson education	2nd Ed. 2022

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

PRODUCTION ENGINEERING

SEMESTER S4

MATHEMATICS FOR PHYSICAL SCIENCE – 4

Course Code	GCMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus.	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To provide the students with the basics of various numerical methods to develop problem solving skills used in various engineering disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random	9

	variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	
3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t -distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Newton-Raphson Method, Gauss Elimination Method, Gauss - Jordan Method, Numerical solution of ordinary differential equations-Euler's method, Modified Euler's method, Runge - Kutta method of 2 nd Order, Numerical solution of Laplace equation –Jacobi's Method, Curve Fitting by Method of Least Squares - Straight lines, Parabola. (Text 2: Relevant topics from sections 2.5, 4.2, 7.5, 8.4, 8.5, 9.4)	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z -tests and the one-sample t -test.	K3
CO4	Apply numerical methods to find solutions of linear system of equations, ordinary differential equations and Laplace equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introductory Methods of Numerical Analysis	S S Sastry	PHI Learning Pvt Limited	5 th edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Numerical methods for Engineers	Steven C. Chapra, Raymond P. Canale	McGraw Hill Education	8 th edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/111/107/111107105/

SEMESTER S4

MANUFACTURING PROCESSES

Course Code	PCPET402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of various manufacturing processes, including casting, forming, machining, joining, and additive manufacturing.
2. To enhance students' problem-solving skills by applying engineering principles to address real-world manufacturing challenges.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of casting processes, Advantages and limitations of casting, Classification of casting processes, solidification. Mold Design and Gating Systems ,Principles of mould design considerations for moulds and cores ,Gating systems: types and design, Riser design and solidification time calculation, Use of chills and chaplets. Types of sand moulds and their properties ,Properties of moulding sands, Pattern making and types of patterns, Core making and core materials, Steps in the sand casting process, Advantages and disadvantages of sand casting	11

	Special Casting Processes : Investment casting (lost wax process),Die casting, Centrifugal casting, Shell moulding, Continuous casting, Squeeze casting Casting defects: Porosity, Shrinkage ,Inclusions Cold shuts ,Misruns, Hot tears, Surface cracks ,Blowholes, Sand inclusions, Runout	
2	Metal joining: classification; welding heat sources; fusion and non-fusion welding processes; gas welding; electric arc welding; arc welding machines; physics of welding arc, polarity; arc production; arc characteristics; metal transfer; arc blow; welding electrode; consumable and non-consumable arc welding, SMAW; GTAW; GMAW; SAW; AHW; PAW; resistance welding; electro slag welding; thermit welding; solid state welding; Welding metallurgy: Heat Affected Zone; weldability of ferrous and non-ferrous metals; design of weldments; joint design; residual stresses and distortion; defects in welding; testing of welded joints; brazing and soldering, beam welding; solid state welding; welding metallurgy: Heat Affected Zone; weldability of ferrous and non-ferrous metals; residual stresses and distortion; defects in welding; testing of welded joints; brazing ,soldering and braze welding.	11
3	Powder metallurgy: Introduction to powder metallurgy, Applications, powder fabrication methods, metallic and ceramic powders, microstructural control, powder characterization, powder treatment: mixing granulation, degassing, coating; Powder compaction and sintering. Principles of sintering: solid state sintering, liquid phase sintering mechanisms: diffusion, vapor phase transport, viscous flow Sintering atmospheres and their effects Secondary operations: machining, heat treatment, surface finishing Hot isostatic pressing (HIP) Infiltration and impregnation Joining techniques for powder metallurgy components	11
4	Advanced manufacturing process: Abrasive jet machining (AJM),Water jet machining (WJM),Abrasive water jet machining(AWJM),Electrochemical	11

	machining (ECM),Electro discharge machining (EDM), Electron beam machining (EBM)and electron beam welding(EBW),Laser beam machining (LBM)and processes, laser beam welding (LBW),ultrasonic welding (USW)	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Demonstrate a fundamental understanding of various manufacturing processes such as casting, forming, machining, joining, and additive manufacturing.	K3
CO2	Evaluate and select appropriate manufacturing processes for different materials and product designs.	K3
CO3	Analyze and solve manufacturing-related problems by applying engineering principles and techniques.	K3
CO4	Operate and handle manufacturing equipment and tools safely and effectively.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	2	-	-	-	-	2
CO2	3	3	3	-	-	-	2	-	-	-	-	2
CO3	3	3	3	-	-	-	2	-	-	-	-	2
CO4	3	3	3	-	-	-	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Modern Manufacturing: Materials, Processes, and Systems	Mikell P. Groover	Wiley	5th Ed. 2012
2	Manufacturing Engineering and Technology	Serope Kalpakjian and Steven Schmid	Pearson	7th Ed. 2017
3	Manufacturing Science	Ghosh and Mallik	East West Press.	2nd Ed. 2010
4	Manufacturing Technology: Foundry, Forming and Welding	P.N. Rao	McGraw Hill Education	4th Ed. 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Materials and Processes in Manufacturing,	DeGarmo, Black, and Kohser	Wiley	3rd Ed. 2017
2	Product Design for Manufacture and Assembly	G. Boothroyd, P. Dewhurst, and W. Knight	CRC Press	3rd Ed. 2010
3	Welding Engineering and Technology	Dr. R.S. Parmar	McGraw Hill Education	4th Ed. 2017

SEMESTER S4

MECHANICS OF METAL CUTTING AND MACHINE TOOLS

Course Code	PCPET403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

- 1.To enable students to familiarize the basics of mechanics of metal cutting.
- 2.To familiarize with working of standard machine tools and their applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Generating Motions of Machine Tools -Machines Using Single-Point Tools -Machines Using Multipoint Tools-ASA and ORS Tool Nomenclatures- Machining Using Abrasive Wheels-Machine Tool Characteristics and Machining Equations. Mechanics of Metal Cutting: Chip Formation – Merchant’s theory-The Forces Acting on the Cutting Tool and Their Measurement -Specific Cutting Energy -Plowing Force and the ‘Size Effect’ -The Apparent Mean Shear Strength of the Work Material -Chip Thickness - Friction in Metal Cutting. Temperatures in Metal Cutting: Heat generation-Temperature Distribution in Metal Cutting -The Measurement of Cutting Temperatures.	11
2	Tool Life and Tool Wear: Progressive Tool Wear-Forms of Wear in Metal	11

	Cutting-Taylor's tool life Criteria-The Tool Material-The Work Material – Machinability Index. Cutting Fluids and Surface Roughness: Cutting Fluids-The Action of Coolants and Lubricants -Application of Cutting Fluids -Surface Roughness ,chip control, chip breakers, chatter, hot machining. Economics of Metal-Cutting Operations: Choice of Feed-Choice of Cutting Speed -Tool Life for Minimum Cost and Minimum Production Time.	
3	Centre Lathe: Construction- Various Operations, Taper Turning Methods, Thread Cutting operation, Lathe Attachments & Accessories. Types of Automation-Transfer Machines -Automatic Machines-Numerically Controlled (NC) Machine Tools-Comparison of the Economics of Various Automation Systems -Flexible Manufacturing System. Reciprocating Machine Tools-Shaper: Principal parts, Classification, Specification of shaper, Shaper Mechanisms, Hydraulic shaper. Cutting Speed, Feed, Depth of cut & machining time-Shaper Operations-Shaping of guide ways. Introduction to Planer -Principal parts and working of Double Planer, Planing of Guide gibs- Slotter-Principal parts of Slotter-Working of slotter-Slotting of Keyways .	11
4	Drilling operations: Twist drill geometry–Radial drilling machine-calculation of machining time- Boring Machine-features-Operations-Counter boring-Spot facing, Countersinking, Trepanning, Reaming, Tapping. Milling: Classification, Column and knee type milling machine - Milling cutters and Classification-Fundamentals of milling Processes-Milling operations. Cutting speed, feed, depth of cut and machining time. Indexing Methods-Simple and compound indexing. Milling of Keyways. Gear Cutting -shaping, forming and generation(Gear hobbing), Gear broaching, Gear Shaving. Abrasive Processes: Grinding wheel-Designation and Selection-Types of Grinding Operations-Honing, Lapping, other finishing processes.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of metal cutting, including chip formation, Merchant's theory, and forces acting on the cutting tool, and apply this knowledge to optimize cutting parameters	K3
CO2	Analyze the factors affecting tool life and tool wear, including Taylor's tool life criteria, and apply that to evaluate the economics of metal-cutting operations to select optimal cutting conditions	K3
CO3	Describe the construction and operations of various machine tools , their selection as per application as well as get insight of the economics of different automation systems	K2
CO4	Assimilate various aspects of machining processes such as drilling, boring, milling, and gear cutting operations and select these based on the specific surface finish and precision requirements	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	-	-	-	-	-	-	-	1
CO2	3	3	2	1	-	-	-	-	-	-	-	1
CO3	2	2	2	-	-	-	-	-	-	-	-	1
CO4	2	2	2	-	-	-	-	-	-	-	-	1
CO5	2	2	2	-	-	-	-	-	-	-	-	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Machining and Machine tools	AB Chattopadhyay	Wiley India	2nd Ed. 2017
2	Fundamentals of machining and machine tools	Geoffrey Boothroyd, Winston A Knight	Taylor & Francis India	3rd Ed. 2005
3	All about machine tools	Heinrich Gerling	New Age International	2nd Ed. 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Technology, Vol II- Metal Cutting and Machine Tools	Rao P N	McGraw-Hill India	3rd Ed. 2013
2	Fundamentals of Metal cutting and Machine tools	B L Juneja, G S Sekhson	New Age International	3rd Ed. 2003
3	Metal cutting principles	M C Shaw	Oxford University Press	2nd Ed. 2012
4	Machining Technology and Operations- Volume-1 Traditional machining technology	Helmi Youssef, Hassan El-Hofy	Taylor & Francis	2nd Ed. 2022
5	Computer Control of Manufacturing System	Yoram Koren	McGraw-Hill India	1st Ed. 2017
6	Automation, Production Systems and Computer Integrated Manufacturing	Mikell P Groover	Pearson	5th Ed. 2021

Video Links	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105233/
	Lecture Video 1-9 (Tool Geometry, Mechanics of Chip formation, Cutting Force) https://www.youtube.com/watch?v=ljveGnQw2G0
2	Metal Cutting and Machine Tools – IITKGP (Prof.Asimava Roy Choudhury) Lecture Video 11-12 (Tool Wear, Tool life https://www.youtube.com/watch?v=BvMrxFCd-3Y
	Introduction to Machining and Machining Fluids-IITG (Dr M Ravi Sankar) Tool wear and Tool life https://www.youtube.com/watch?v=cE-GPE6HtqM
	Introduction to Machining and Machining Fluids-IITG (Dr M Ravi Sankar) https://www.youtube.com/watch?v=m9Ku8JLhlg8&list=PLY2p6n3OIhu6trgUqFIIBr2AkqOyUN5bj&index=1
3	Metal Cutting and Machine Tools – IITKGP (Prof.Asimava Roy Choudhury) Lecture Video 13-15 (Lathe operations) https://www.youtube.com/watch?v=1AwOzDv7j6o
4	Metal Cutting and Machine Tools – IITKGP (Prof.Asimava Roy Choudhury) Lecture Video 16-18 (Milling Machine, Gear cutting) https://www.youtube.com/watch?v=sKEw48dkB90
	Working and Operation with Indexing Head https://www.youtube.com/watch?v=Rf90Jbbcr3M
	Spur and Helical Gear Cutting—IITKGP (Prof.Asimava Roy Choudhury) Gear Shaping and Gear Hobbing https://www.youtube.com/watch?v=QRbFsNUMZ54&list=PLn6IrNKKOidnzCHfk7u-emfuz4h1UdTqF&index=11

SEMESTER S4

MECHANICS OF MACHINERY AND ROBOTICS

Course Code	PBPET404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize different mechanisms and enable the students to do kinematic analysis.
2. To equip the students to understand and apply Robot kinematics and Robot dynamics in real world situations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to kinematics of Machinery: Linkages and joints -kinematic pairs and chains-degrees of freedom and mobility-Inversion of 4 bar chain and its various applications. Kinematic Analysis: Displacement analysis, velocity, and acceleration analysis (includes Coriolis acceleration)-Graphical approach. Velocity analysis by Instantaneous centre method.	10
2	Introduction to kinematic synthesis: Type synthesis and Dimensional synthesis-Function generation -Path generation- Motion generation-Chebyshev's spacing of precision points. Graphical method for dimensional synthesis: 2 position and 3 position synthesis. Analytical Method-Freudenstein's equation- loop closure	10

	equation. Synthesis of lower pair mechanisms such as Straight line and quick return mechanisms; Higher Pairs: Cam and follower for prescribed displacement - Gears and gear trains for prescribed speed and torque transmission.	
3	Introduction to Robotics-Industrial robot manipulators, Service applications, electro mechanical components. Robotic kinematics:Kinematic models of manipulators, definition of pose-position description-Orientation description, changing description from frame to frame, homogeneous transformation matrix. Manipulator Kinematics: link description and link to reference frame connection - Denavit-Hartenberg (DH) representation-Forward kinematics, Inverse Kinematics-Formulation - Enumerated using simple manipulator configurations. Workspace Analysis and Trajectory Planning-planning with constraints (Case study-based explanation only).	12
4	Velocity analysis and Static Analysis of manipulators: Linear and angular velocity of links-Jacobian matrix, singularity, Static Analysis of manipulator. Robot Dynamics: Introduction to dynamics, Lagrangian formulation (Overview)of manipulator dynamics. Linear robot dynamics -Force acceleration relationship for 1 DOF system, equation of motion for 2 DOF robot arm. Motion Control:Path v/s Trajectory, Joint space v/s cartesian description, Point to point control, trajectory generation, continuous path control, force control, hybrid position/force control system. Overview of Robot Programming, scope for mobile robots, need and application of AI and ML in Robotics.	12

Suggestion on Project Topics:

- Design and Analysis of a Four-Bar Linkage for a Specific Application.
- Velocity analysis of quick return mechanism by Instantaneous centre method.
- Synthesis of a Cam Mechanism for a given Motion Program.

- Design of a Mechanism for Converting Rotational Motion to Linear Motion.
- Analysis of a Pantograph Mechanism for a specific task.
- Develop a program to generate DH parameters for a Manipulator.
- Design of a Robot Arm for Pick-and-Place Tasks.
- Forward and Inverse Kinematics Analysis of a 2-DOF Robot Arm.
- Development of a robot arm for a pick-and-place task in a manufacturing line.
- Kinematic synthesis of a mechanism for a robotic arm for a specific industry.
- Design of a Robot Gripper for a specific object manipulation task.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 subdivisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Identify and appreciate the key Mechanisms they encounter in real world and engineering scenarios.	K3
CO2	Apply the core concept of kinematics for the design and synthesis of single degree of freedom of mechanisms.	K3
CO3	Design and synthesis of robotics mechanisms for achieving desired output motion.	K5
CO4	Evaluate and select appropriate mechanisms and robotics systems for specific applications.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	1	-	-	-	-	-	1
CO2	3	3	2	1	-	1	-	-	-	-	-	1
CO3	3	3	3	2	2	-	-	-	-	-	-	2
CO4	3	3	3	1	2	-	-	-	2	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of Machines	S S Rattan	McGraw Hill Education	4th Ed. 2017
2	Theory of Machines and Mechanisms	P L Ballaney	Khanna Publishers	25th Ed. 2022
3	Introduction to Robotics	Saha S K	McGraw Hill Education	2nd Ed. 2017
4	Modern Robotics: Mechanics, Planning, and Control	Frank Chongwoo Park, Kevin M. Lynch	Cambridge University Press	1st Ed. 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanism and Machine Theory	Ashok G. Ambekar	Prentice Hall India	1st Ed. 2007
2	Mechanism Design-Analysis and Synthesis. Vol-1	Arthur G.Erdman, George N.Sandor, Sridhar Kota	Pearson Education	4th Ed. 2001
3	Introduction to Robotics Mechanics and Control	John Craig	Pearson Education,	4th Ed. 2022
4	Kinematics of Machinery through HyperWorks	J.S.Rao	Springer	1st Ed. 2011
5	Theory of Machine and Mechanisms	John J.Uicker,Gordon R. Pennock,JosephE.Shigley	Oxford University Press	5th Ed. 2017
6	Robotics Technology and Flexible Automation	Deb S R.andDebS	McGraw Hill Education (Ind)	2nd Ed. 2010
7	Robotics: Designing the mechanism for Automated Machinery	Ben-Zion Sandler	Academic Press	2nd Ed. 1999

Video Links	
Module No.	Link ID
1	Kinematics of Machines – Prof. A. K. Mallik (fundamentals concepts) https://nptel.ac.in/courses/112/104/112104121/
	Kinematics of Mechanism and Machines -Prof. A. Das Gupta (displacement, velocity and acceleration Analysis) https://nptel.ac.in/courses/112/105/112105268/
2	Theory of Mechanism- Prof.SujathaSrinivasan(Mechanism Synthesis) https://archive.nptel.ac.in/courses/112/106/112106270/
3	Mechanics and Control of Robotic Manipulators (NPTEL/NOC/IITPkd)-Prof.Dr.Santhakumar Mohan https://archive.nptel.ac.in/courses/112/106/112106304/
4	Mechanics and Control of Robotic Manipulators (NPTEL/NOC/IITPkd)-Prof.Dr.Santhakumar Mohan https://archive.nptel.ac.in/courses/112/106/112106304/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

MACHINE DESIGN

Course Code	PEPET411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objective:

- To familiarize and apply the principles of stress strain and deformation to analyse and solve problems in mechanical components and structures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Common Engineering materials and its Properties, Selection of Materials, steps in design Process Principles of standardisation, limit and fits as per IS specification. Estimation of design loads, factor of safety, Theories of failure, Stress concentration factor, Variable stress, fatigue failure, endurance limit, design for finite and infinite life, Soderberg and Goodman criteria.	9
2	Detachable joints: design of screws, standards, thread stresses, preloading of bolts, fatigue and shock load, eccentric loading. Power screws, mechanism of power screws, thread stresses, efficiency of power screws. Riveted joint: Stresses in riveted joint, design of riveted joints with central and eccentric loads, boiler and tank joints, structural joints.	9

	Welded joints: types of welded joints, stresses, design of welded joints subjected to axial, torsional and bending loads, welds subjected to fluctuating loads.	
3	Springs: Classification, Common Spring materials, stress and deflection in helical compression spring with axial loading, curvature effects, resilience, spring surging, buckling, critical frequency of helical springs – helical torsion springs design of automotive leaf springs. Pressure vessels, thin cylinders, Thick cylinder equation, Hydraulic accumulators. Compound Cylinders.	9
4	Design of gears: design of spur, helical, bevel gears, virtual or formative number of teeth - gear tooth failures- beam strength, Lewis equation, Buckingham's equation for dynamic load, wear load, endurance strength of tooth, heat dissipation. Design of Sliding contact bearings: lubricants & viscosity - journal bearings, hydrodynamic theory, bearing characteristic number, design considerations. Design of Rolling contact bearing: Classification, bearing life, static and dynamic load capacity, dynamic equivalent load, selection of bearings.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand mechanical design procedure, Material selection, Codes & use of standards.	K2
CO2	Apply the knowledge in designing detachable and permanent joints.	K3
CO3	Determine safe design of spring and Pressure vessels describe different types of bearings and clutches, its application; design procedures.	K3
CO4	Describe different types of bearings and Gears, its application; design procedures.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	2	-	-	-	2	2
CO2	3	3	3	-	-	-	2	-	-	-	2	2
CO3	3	3	3	-	-	-	2	-	-	-	2	3
CO4	3	3	3	-	-	-	2	-	-	-	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of machine Elements	V B Bhandari	McGraw Hill Education	5th Ed. 2020
2	Design of machine Elements	M. F. Spotts & Terry E. Shoup	Pearson Education	8th Ed. 2019
3	Machine Component Design	Robert C. Juvinall & Kurt M. Marshek	WILEY	5th Ed. 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Machine Elements	Dr.G.K.Vijayaraghavan	Lakshmi Publications	6th Ed. 2015
2	Shigley'S Mechanical Engineering Design	Richard G. Budynas J. Keith Nisbett	McGraw Hill	11th Ed. 2020

SEMESTER S4

MATERIAL CHARACTERIZATION TECHNIQUES

Course Code	PEPET412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the methods required to characterize materials such as metals and alloys.
2. To prove a comprehensive understanding of characterizing materials based on various properties.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to material characterization techniques, Structure analysis tools: X-ray diffraction, Phase identification, indexing and lattice parameter determination. Analytical line profile fitting using various models, Neutron diffraction; Reflection High energy electron Diffraction (RHEED), Low energy Electron Diffraction (LEED).	9
2	Microscopy techniques: Introduction to Microscopes, Optical Microscopy (OM), Transmission Electron Microscopy (TEM); Basic Electron scattering, Concepts of resolution. TEM instruments, Various imaging modes, Analysis of micrographs, Electron Energy Loss Spectroscopy, Scanning Electron Microscopy, Rutherford backscattering spectrometry.	9

3	Atomic Force Microscopy, Scanning Probe Microscopy, Thermal analysis techniques: Differential Thermal Analysis (DTA), Differential Scanning Calorimetry (DSC), Thermo-gravimetric analysis (TGA), Electrical Characterization Techniques: Electrical resistivity in bulk and thin films, Hall effect, Magnetoresistance.	9
4	Magnetic characterization techniques: Introduction to Magnetism, Measurement Methods, Measuring Magnetization by Force, Measuring Magnetization by Induction method, Types of measurements using magnetometers: M-H loop, temperature dependent magnetization, time dependent magnetization. Measurements using AC susceptibility, Magneto-optical Kerr effect, Nuclear Magnetic Resonance, Electron Spin Resonance, Optical characterization techniques: UV-VIS spectroscopy, Fourier transform infrared spectroscopy, Raman spectroscopy, X-ray photoelectron spectroscopy.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Summarize about tools used for structure analysis of crystals	K2
CO2	Explain various microscopy techniques used in material characterization.	K2
CO3	Classify thermal and electrical characterization techniques.	K2
CO4	Outline about magnetic and optical characterization techniques	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	2	3	-	-	-	-	-	-	3
CO2	3	2	-	2	2	-	-	-	-	-	-	3
CO3	3	3	-	3	2	-	-	-	-	-	-	3
CO4	3	3	-	3	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Materials Characterization Techniques	S Zhang, L. Li and Ashok Kumar	CRC Press	1st Ed. 2008.
2	Materials Science and Technology: A Comprehensive Treatment, Vol 2A & 2B)	Robert Wolfgang Cahn, Peter Haasen, Edward J. Kramer, Eric Lifshin	VCH	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Physical methods for Materials Characterization	P. E. J. Flewitt and R K Wild	IOP Publishing	3rd Ed. 2017
2	Semiconductor Material and Device Characterization	D. K. Schroder	Wiley-IEEE Press	3rd Ed. 2006
3	Characterization of Nanophase Materials	Ed. Z L Wang	Willet-VCH	1st Ed. 2000
4	Materials Characterisation, Vol. 10	ASM hand book	ASM hand book	9th Ed., 1997
5	ASTM handbook, Vol. 3	ASTM hand book	ASTM hand book	1997
6	Elements of X-ray diffraction	B.D. Cullity, S. R. Stock	Pearson Education Ltd.	3rd Ed. 2014
7	Thermal Analysis of Materials	R.F. Speyer	CRC Press	eBook 2014

Video Links	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/106/112106227/
2	https://onlinecourses.nptel.ac.in/noc22_mm14/preview
3	https://nptel.ac.in/courses/113105100
4	https://archive.nptel.ac.in/courses/115/103/115103030/

SEMESTER S4

MACHINE LEARNING

Course Code	PEPET413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To learn and implement various machine learning algorithms.
2. To gain hands-on experience with machine learning frameworks and tools.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Python basics: Data types, control structures, functions, Introduction to Object-Oriented Programming (OOP), Classes and objects, inheritance, polymorphism, encapsulation. Advanced Python concepts: List comprehensions, lambda functions, decorators Introduction to NumPy , Scikit-learn and pandas for data manipulation, Hands-on with NumPy arrays and pandas Data Frames.	9
2	Recap on Linear Algebra Essentials: Vectors, matrices, and operations Eigenvalues and eigenvectors, Linear transformations. Calculus for Machine Learning: Differentiation and integration, Partial derivatives and gradients, Applications in optimization and learning algorithms. Probability theory: Basic concepts, distributions, Descriptive statistics: Mean, median, variance, Inferential statistics: Hypothesis testing, confidence intervals.	9

	Data cleaning: Handling missing values, outliers, Data normalization and scaling Exploratory Data Analysis (EDA)	
3	Introduction to Supervised Learning: Linear regression: Simple and multiple linear regression. Classification: Logistic regression, decision trees, k-nearest neighbours (KNN). Performance metrics: Accuracy, precision, recall, F1-score. Introduction to Unsupervised Learning: Clustering: K-means, hierarchical clustering, DBSCAN. Dimensionality reduction: Principal Component Analysis (PCA), t-SNE.	9
4	Model Evaluation and Hyperparameter Tuning: Model evaluation techniques: Train-test split, cross-validation, Hyperparameter tuning: Grid search, random search. Advanced Machine Learning Algorithms :Support Vector Machines (SVM), Ensemble methods: Bagging, boosting, random forests, Neural networks and introduction to deep learning. Case studies on machine learning applications in various domains.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand the principles and concepts of machine learning.	K2
CO2	Implement basic machine learning algorithms.	K3
CO3	Apply machine learning models to solve practical problems.	K3
CO4	Utilize machine learning frameworks and tools for developing and testing models.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	3	-	2	-	-	-	-	2
CO2	3	3	3	2	3	-	2	1	-	-	-	2
CO3	3	3	3	3	3	-	2	1	-	-	-	3
CO4	3	3	3	3	3	-	2	1	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Machine Learning	Tom Mitchell	McGraw-Hill	1st Ed. 2017
2	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow	Aurelien Geron	O'Reilly Media	3rd Ed. 2020
3	Mathematics For Machine Learning	Marc Peter Deisenroth.	Cambridge University Press	1st Ed. 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep Learning	Ian Goodfellow, Yoshua Bengio	MIT Press	1st Ed. 2016
2	Python Machine Learning	Sebastian Raschka and Vahid Mirjalili:	Packt Publishing	2nd Ed. 2017

Video Links	
Link ID	
https://www.youtube.com/watch?v=jGwO_UgTS7I&list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU	

SEMESTER S4

MATERIAL HANDLING AND MATERIALS MANAGEMENT

Course Code	PEPET414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize techniques required for material handling and materials management.
2. The course discusses various methods and approaches for materials managing and handling in various manufacturing and service industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to materials handling techniques- Definitions, scope and objectives- Traditional and modern views of material handling. Principles of material handling-Concept of unit load- Material handling classification-by materials-by method-by equipment type-Stages in material handling system design.	9
2	Introduction to material storage and store keeping-Scope and objectives of store keeping-Functions of store keeping-Features of successful store keeping-Relationship of stores department with other departments-Benefits of stores keeping-Stores Organisation. Deciding the location and layout of stores- Classification and codification of materials-merits and demerits- Types of stores and storage equipment - Preservation of stores and stock verification.	9
3	Introduction to materials management-Scope, primary and secondary objectives of materials management-Functions of management as applied to	9

	materials -Addressing the conflicting objectives in materials management. Role of specifications in purchasing and materials management- development of simple and complex specifications and common problems with specifications-Budgeting and materials planning .Purchasing cycle and principles of purchasing- Value analysis and techniques for materials management.	
4	Introduction to integrated materials management- Need for integrated concept- Advantages of integrated materials management -Organisation for integrated materials management-Activities of materials management. Desirable qualities of Purchasing and materials manager-Interdepartmental relationship in to integrated materials management- Performance appraisal of materials management- Management Information system for materials management.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Summarize techniques for material handling.	K2
CO2	Explain various techniques used for material storage.	K2
CO3	Explain various functions for materials management.	K2
CO4	Outline the concept of integrated materials management.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	2	3	-	-	-	-	-	-	3
CO2	3	2	-	2	2	-	-	-	-	-	-	3
CO3	3	2	-	3	2	-	-	-	-	-	-	3
CO4	3	2	-	3	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Materials Management- Text and Cases	A K Chitale, R C Gupta	PHI Learning Private Ltd.	2nd Ed. 2011
2	Materials Management- Procedure, Text and cases	A K Datta	PHI Learning Private Ltd.	2nd Ed. 2005
3	Introduction to Materials Management	J. R. Tony Arnold, Stephen N. Chapman, Lloyd M. Clive	Pearson Education	7th Ed. 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Materials Management: An Integrated Systems Approach	P. Gopalakrishnan, M. Sundaresan	Prentice-Hall of India Pvt. Ltd.	1st Ed. 2004
2	Materials Handling Handbook	John Wiley & Sons	John Wiley & Sons	2nd Ed. 1985

SEMESTER S4

STATISTICAL QUALITY CONTROL

Course Code	PEPET415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize statistical quality control methods to monitor and improve engineering processes.
2. To enable students to practice effective statistical quality control methods for solving real-world problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic Concepts of Quality: Meaning of quality, quality of design, quality of conformance, quality of performance, quality function, goal post view of quality, other aspects of quality, quality and profitability, quality control, quality characteristics, cost of quality, optimum cost of performance, value of quality, balance between cost of quality and value of quality, specification of quality, quality control and inspection, quality policy, statistical quality control (SQC), role of feedback in quality control, quality perspective. Quality Assurance: Quality assurance manual, field complaints, quality rating of outgoing product, quality survey or quality audit, Executive report on quality, inspection planning, quality budget, vendor quality rating, organization for quality function, organization for acceptance, manufacturing planning for quality, planning for process control, statistical process control (SPC), Quality function deployment (QFD), quality policy deployment.	9

2	Total Quality Management: Total quality control, need for management of product quality, concept of total quality management(TQM), TQM philosophies, Taguchi philosophy, concept of loss function, orthogonal arrays, linear graphs, analysis of variance(ANOVA), signal to noise(S/N) ratio, Crosby's fourteen steps, TQM models, Training for quality management, ISO:9000 and TQM, benchmarking, Kaizen and Kairyo systems, tools and techniques of Kaizen, brainstorming, DMAIC cycle, Poka-Yoke, JIDOKA, waste control, 5S for improvement, Just in time manufacturing (JIT), total quality management for excellence (TQMEX).	9
3	Control Charts for Variables: Concept of variation, distinction between variables and attribute data, theory of control charts, definition of control chart, relationship between $\bar{X}'$, σ' and the values of $\bar{X}$, relationship between σ' and $\bar{\sigma}$, relationship between σ' and $\bar{R}$, choice of variables, basics of sub-grouping, size and frequency of subgroups, control limits, chance of making an error, starting the control charts, conclusions from control charts, control chart patterns, run sum test, revising control limits, process capability analysis. Control Chart for Attributes: Limitations of control chart for variables, Attribute charts, comparison of $\bar{X}$ and R with p chart, control limits on p chart, choice between p chart and np chart, periodic review and revision of p chart, control charts for defects	9
4	Acceptance Sampling: Introduction, sampling methods, operating characteristic curve (OCC), producer's risk and consumer's risk, quality indices for acceptance sampling plans, Average outgoing quality limit (AOQL), steps in the design of an acceptance plan, sampling plans, Average total inspection (ATI) curve, Dodge –Romig system. Use Excel/SPSS to plot control charts and to solve problems in sampling.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Examination</i>	<i>Analyse</i>	<i>Evaluate</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> <i>a. Clearly defines the real-world quality issue..</i> <i>b. Examine and identify relevant contextual factors.</i>	5
	<i>Problem Analysis</i> <i>a. Present a structured realistic solution methodology.</i> <i>b. Compare and justify the proposed solutions with evidence and logical reasoning</i>	5
Evaluate	<i>Validation of Results</i> <i>a. Thoroughly evaluate the proposed solutions.</i> <i>b. Compares trade-offs, advantages, and disadvantages.</i> <i>c. Considers feasibility, scalability, and practical implications</i>	5
	<i>Conclusion and Report Writing</i> <i>a. Summarizes procedure, findings and insights, limitation and scope for future work.</i> <i>b. Preparation of report with all components of project report.</i>	5

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
- Minimum 1 and Maximum 2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 subdivisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand the basics of quality and quality assurance.	K2
CO2	Understand the basics of TQM.	K2
CO3	Apply principles and models of control charts to monitor and control the engineering processes and to analyze the capabilities of the processes.	K4
CO4	Apply and analyze acceptance sampling techniques to solve real world problems.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	2	-	-	2
CO2	3	2	1	1	-	-	-	-	2	-	-	2
CO3	3	3	2	1	2	-	-	-	2	2	-	2
CO4	3	3	2	2	3	-	-	-	2	2	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Quality Control	M. Mahajan	Dhanpat Rai & Co	Revised Ed. 2016
2	Introduction to Statistical Quality Control	D. C. Montgomery	John Wiley Sons	7th Ed. 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Quality Control and Improvement	Mitra A	Wiley	4th Ed. 2016
2	Statistical Quality Control	Eugene L Grant and Richard S Leavenworth	McGraw-Hill Education	7th Ed. 2017
3	Statistical Quality Control	M Jeya Chandra	CRC Press LLC	1st Ed. 2001

Video Links	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/110/105/110105088/
2	https://archive.nptel.ac.in/courses/110/105/110105088/
3	https://archive.nptel.ac.in/courses/110/105/110105088/
	https://archive.nptel.ac.in/courses/116/102/116102019/#
4	https://archive.nptel.ac.in/courses/110/105/110105088/
	https://archive.nptel.ac.in/courses/116/102/116102019/#

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		<i>Answer all questions. Each question carries 3 marks</i>	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
<i>Answer any one full question from each module. Each question carries 8 marks</i>				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of	6

	confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable	6

	Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	6

**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.

- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group /Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.

- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).

- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

MACHINE TOOL LAB

Course Code	PCPEL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To impart hands-on training on conventional and advanced machine tools.
2. To provide training in precision machining operations with adherence to the standards

Expt. No.	Experiments (Mandatory)
1	Step turning, grooving- Part I.
2	Step turning, grooving- Part II.
3	Taper turning.
4	Thread cutting (single start Only).
5	Spur Gear cutting using Simple indexing Method -Part I.
6	Spur Gear cutting using Simple indexing Method -Part II.
7	Shaping rectangular block.
8	Rectangular slot and Dovetail cutting -Part I.
9	Rectangular slot and Dovetail cutting -Part II.
10	Slotting of Keyway.

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva Voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Demonstrate the working principles and applications of various machine tools.	K2
CO2	Make use of machine tools and machine tools operations.	K3
CO3	Utilize different types of cutting tools and their applications in various machining processes	K3
CO4	Analyze and interpret experimental data related to machining processes	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	2	-	-	-	-	3	1	-	3
CO2	3	2	-	2	-	-	-	-	3	1	-	3
CO3	3	2	-	3	-	-	-	-	3	1	-	3
CO4	3	3	-	2	-	-	-	-	3	1	-	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Production Technology: Manufacturing Processes, Technology, and Automation	R.K. Jain and S.C. Gupta	Khanna Publishers	1st Ed. 2001
2	Manufacturing Processes for Engineering Materials	Serope Kalpakjian and Steven Schmid	Pearson Education	6th Ed. 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Science	Ghosh and Mallik	East West Press.	2nd Ed. 2010
2	Manufacturing Technology: Foundry, Forming and Welding	P.N. Rao	McGraw Hill Education	4th Ed. 2017

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

PRODUCTION ENGINEERING LAB

Course Code	PCPEL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To provide students with a solid foundation in the principles and practices of various manufacturing processes.
2. To equip students with hands-on experience in operating production machinery and conducting manufacturing experiments.

Expt. No.	Experiments (Minimum 10 Experiments should be completed)
1	Testing of Green compression strength.
2	Hot compression strength.
3	Sand Permeability test.
4	Sieve Analysis -determination of AFS number.
5	Flowability measurement of Metals.
6	Shear angle measurement in Orthogonal metal cutting.
7	Weld Strength test.
8	Study of Heat Affected Zone(HAZ) in welding -Metallurgical study and Micro hardness test.
9	Formability test of sheet metals using Ericsson Cupping Test.
10	Deformation Analysis of sheet metals.
12	Tool wear assessment and verification of Taylor's Tool life equation.
12	2 axis and 3 axis cutting Forces measurement.
13	Non Destructive Testing of Components -Dye Penetrant test, Magnetic Crack detection.
14	Acceptance Test of Machine tools.
15	Eccentric turning of a component in Lathe.
16	Multi start thread cutting in Lathe.
17	NC Manual part programming and its simulation.

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva Voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Demonstrate a comprehensive understanding of various manufacturing processes such as casting, welding, machining, and forming.	K3
CO2	Utilize production machinery and tools, ensuring safety and efficiency in a manufacturing environment	K3
CO3	Analyze and optimize manufacturing processes to enhance productivity, reduce waste, and improve overall efficiency.	K4
CO4	Utilize problem-solving skills to address and troubleshoot issues encountered in manufacturing processes.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	2	3	-	-	-	2	2	-	3
CO2	3	2	-	2	3	-	-	-	2	2	-	3
CO3	3	2	-	2	2	-	-	-	2	2	-	3
CO4	3	3	-	2	2	-	-	-	2	2	-	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Production Technology: Manufacturing Processes, Technology, and Automation	R.K. Jain and S. C. Gupta	Khanna Publishers	1st Ed. 2001
2	Manufacturing Processes for Engineering Materials	Serope Kalpakjian and Steven Schmid	Pearson Education	6th Ed. 2018

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1	Manufacturing Technology: Foundry, Forming and Welding	P. N. Rao	McGraw Hill Education	4th Ed. 2017
2	Manufacturing Science	Ghosh and Mallik	East West Press.	2nd Ed. 2010

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

PRODUCTION ENGINEERING

SEMESTER S5
INDUSTRIAL ENGINEERING

Course Code	PCPET501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to understand industrial engineering and productivity concepts.
2. To enable students to furnish adequate background for applying concepts of work study, industrial relations and industrial safety.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to industrial engineering: Modern concepts in industrial engineering, functions of industrial engineer, applications of industrial engineering. Productivity: Definition and concepts, input output model, factors affecting productivity, productivity ratios, measurement of productivity, productivity improvement, productivity in Indian industries, productivity and product design process.	11
2	Productivity and work study: Role of work study in improving productivity, specific contributions of method study and work measurement. Method study: Procedure of method study - selection, recording and representation of work and activities, critical analysis, developing best method, installation and maintenance of standard method.	11
3	Work measurement: Purpose of work measurement, basic procedure, performance rating, concept of normal time, allowances. Work sampling technique of work measurement, introduction to predetermined motion time systems. Motion Study: Memo motion and micro motion study, therbligs, SIMO chart, principles of motion economy related to worker, workplace and tools.	11

4	Industrial relations: Objectives, factors affecting industrial relations, psychological attitudes to work and working conditions, Causes and effects of industrial disputes, methods to settle industrial disputes. Industrial fatigue: Factors contributing to industrial fatigue, effect of fatigue, methods to eliminate industrial fatigue. Industrial safety: Objectives, measures for ensuring industrial safety.	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the principles of industrial engineering and concepts of productivity.	K3
CO2	Make use of the concepts of work study for improving productivity and to develop methods for accomplishing work tasks.	K3
CO3	Apply work measurement techniques in organizations and to establish the standard time.	K3
CO4	Identify the concepts of industrial relations and industrial safety in the real world.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	2	-	-	-	-	-	-	-	-	2
CO2	2	-	2	-	-	-	-	-	-	-	-	2
CO3	2	2	2	-	2	-	-	-	-	-	-	2
CO4	2	-	2	-	-	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering and Management	B Kumar	Khanna Publishers	10th Ed., 2004
2	Industrial Engineering and management	O P Khanna	Danpat Rai publication	5th Ed., 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Industrial Engineering	Avraham Shtub, Yuval Cohen	Taylor and Francis	2nd Ed., 2017
2	Introduction to Work study	ILO	Oxford and IBH Publishing	3rd Ed., 2015
3	Reliability Engineering	L S Srinath	East west Press	4th Ed., 2005
4	Statistical Quality Control	Eugene L Grant, Richard S, Leavenworth	McGraw Hill	7th Ed., 2006

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107249/
	https://archive.nptel.ac.in/courses/112/107/112107142/

SEMESTER S5

SENSORS AND MECHATRONIC SYSTEMS

Course Code	PCPET502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the principles of sensors, its characteristics, and its interfacing with microprocessors/microcontrollers.
2. To enable the students to understand the concept of PLC systems, ladder programming and key elements of Mechatronics systems, its modelling and its analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Sensors and Actuators: Performance terminology of sensors, Displacement, Position & Proximity Sensors, Force, Fluid pressure, Liquid flow sensors, temperature, light sensor, Acceleration and Vibration measurement. MEMS, selection of sensor. Mechanical Actuation System, Hydraulic & Pneumatic Actuation System, Electrical Actuation System, Data Presentation system.	11
2	Signal Conditioning: Basic Digital circuits, Introduction to signal processing & Op-Amp, Op-Amp as signal conditioner, Analogue to Digital Converter, Digital to Analogue Converter, filtering, multiplexer, data acquisition. Mechatronics Systems: Elements of mechatronic systems, concept of controls, Mechanical system model, Electrical system model, Fluid system model. Microprocessor, Micro Controller, Introduction to Programming of Microcontrollers.	11
3	Dynamic response of systems: first and second order systems, Transfer function; first and second order systems, systems in series and in feedback loop.	11

	Frequency response: Sinusoidal input, Phasors, Bode plot performance specifications and stability. Closed loop controllers: P, I, PID Controllers, Digital Controllers.	
4	Programmable Logic Controllers: Basic structure, programming, Mnemonics, timers, internal relays, and counters, shift register, master, and jump control. Communication systems: networks and protocols. Input/output systems; interfacing, input/output addressing, and interfacing requirements. Simple case studies based on mechatronic systems.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify different types of sensors and actuators and its applicability to a real time problem.	K3
CO2	Apply the concept of signal processing and to identify the key elements of the mechatronics system.	K3
CO3	Perform time and frequency domain analysis, and design closed loop controllers.	K4
CO4	Develop PLC ladder programming and identify the communication requirements in the systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	2	3	-	-	-	-	-	-	3
CO2	3	2	-	2	2	-	-	-	-	-	-	3
CO3	3	3	-	3	2	-	-	-	-	-	-	3
CO4	3	3	-	3	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering	William Bolton	Pearson	7th Ed., 2023
2	Mechatronic system Design	Dedas Shetty, Kolk and Richard	Pearson	2nd Ed., 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechatronic Handbook	Robert H Bishop	CRC press	2nd Ed., 2007
2	Introduction to Mechatronics	D.G. Alciatore & Michael B. Hstand	Tata Mc Graw Hill	4th Ed., 2012
3	Intelligent Mechatronic Systems: Modelling, Control and Diagnosis	R. Merzouki A K, Samantaray P M, Pathak B Ould Bouamama	Springer	1st Ed., 2013
4	Mechatronic Systems Design	Janschek, Klaus	Springer	1st Ed., 2012
5	Mechatronic Systems: Analysis, Design and Implementation	Boukas K, Al-Sunni, Fouad M	Springer	1st Ed., 2011

Video Links (NPTEL, SWAYAM...)	
Module No	Link ID
1	https://archive.nptel.ac.in/courses/108/108/108108147/
2	https://archive.nptel.ac.in/courses/112/107/112107298/
3	https://archive.nptel.ac.in/courses/112/107/112107214/
4	https://archive.nptel.ac.in/courses/107/106/107106081/

SEMESTER S5

MECHANICAL MEASUREMENTS AND QUALITY CONTROL

Course Code	PCPET503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the concepts of measurement techniques and instruments.
2. To enable students to apply the Quality Control Principles.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of metrology: Need for inspection, accuracy and precision, standards of measurements, hysteresis, linearity, resolution of measuring instruments, threshold, drift, zero stability, loading effect and system response. static performance characteristics, Errors and their classification, concept of calibration. Linear Measurement Instruments: vernier calliper, micrometer, combination set, slip gauges, limit gauges, problems on measurements with gauge. Force measurement: load cells, cantilever beams, proving rings, differential transformers, measurement of torque- torsion bar dynamometer. mechanical strain gauges, electrical strain gauges, strain gauge: materials, gauge factors, theory of strain gauges and method of measurement, bridge arrangement, temperature compensation.	9
2	Measurements of gear and thread parameters: Gear tooth terminology, sources of errors in manufacturing of gears, measurement of tooth thickness- Gear tooth vernier, constant chord method, addendum comparator method and base tangent method, measurement of tooth profile- tool maker's microscope, involute tester, measurement of pitch, measurement of run out, lead and backlash checking, measurement of concentricity, alignment of gears.	9

	Errors in threads, screw thread gauges, measurement of elements of the external and internal threads, thread calliper gauges, tachometers.	
3	Surface Metrology: Concepts and terminology, analysis of surface traces, specification of surface texture characteristics, and method of measuring surface finish: stylus system of measurement, stylus probe instruments, wave length, frequency and cut off, other methods for measuring surface roughness- pneumatic method, light interference microscopes, Mecrin instruments. Comparators: Functional requirements, classification- mechanical comparators, mechanical optical comparators, electrical comparators, pneumatic comparators, Coordinate Measuring Machines.	9
4	Quality and Quality control: Quality concepts, corrective and prevention methods for ensuring quality, cost of quality, organization for quality, ISO standards, Deming prize, Malcolm Baldrige quality award, Rajiv Gandhi National Quality Award, quality audits, quality by design, quality culture, quality from the customer perspective, Quality Information Systems, quality planning, quality tools- six-sigma, process capability (numerical problems), quality control charts (numerical problems based on P, $\bar{X}$ & R Chart), strategic quality management.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the concepts of metrology and measuring instruments.	K2
CO2	Apply the measuring instruments for measuring gear and thread parameters.	K3
CO3	Identify the roughness standards and roughness measurements; make use of the concept of comparators.	K3
CO4	Demonstrate the concepts of quality and quality control.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	2	3	-	-	-	-	-	-	3
CO2	3	2	-	2	2	-	-	-	-	-	-	3
CO3	3	3	-	3	2	-	-	-	-	-	-	3
CO4	3	3	-	3	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Metrology and Measurements	N V Raghavendra, L Krishnamurthy	Oxford University Press	1st Ed., 2013
2	Managing for Quality and Performance Excellence	Evans J R and Lindsay W M	South Western Cengage Learning	9th Ed., 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Physics of Metrology	Alex Hebra	Springer Vienna	1st Ed., 2010
2	Mechanical Measurements	S P Venkateshan	Springer International Publishing	2nd Ed., 2021
3	A Textbook Of Engineering Metrology	I C Gupta	Dhanpat Rai Publications	8th Ed., 2015
4	Theory and Design for Mechanical Measurements	Richard S Figliola, Donald E Beasley	Wiley India Pvt. Limited	3rd Ed., 2008
5	Engineering Metrology	R K Jain	Khanna Publishers	22nd Ed., 2022
6	Total Quality Management	P N Mukherjee	Prentice Hall India	1st Ed., 2006
7	Total Quality Management (TQM) - Principles, Methods, and Applications	Sunil Luthra, Dixit Garg, Ashish Agarwal, Sachin K Mangla	CRC Press	1st Ed., 2020
8	Strategic Continuous Process Improvement	Gerhard J Plenert	McGraw Hill LLC	1st Ed., 2011
9	Total Quality Management	Dale H Basterfield, Glen H Besterfield, Hemant Urdhwareshe	Pearson	5th Ed., 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/106/112106179/
	https://archive.nptel.ac.in/courses/110/104/110104080/

SEMESTER S5

CAD/CAM/CAE

Course Code	PBPET504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the students about the principles and concepts in CAD/CAM and CAE.
2. To enable the students to design and optimize products and manufacturing processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Morphology of design: Sequential and concurrent engineering, manufacturing planning and control, manufacturing models, elements of CAD and CAM, CAD/CAM integration, CAD/CAM data bases and system architecture, requirements of CAD/CAM Packages. Transformations: Geometric and viewing transformation, basic transformation matrices- translation, rotation, scaling etc. Geometric modelling: Mathematical representation of curves, parametric representation of analytical and synthetic curves, parametric equations- circle, ellipse, parabola, synthetic curves, geometric construction of curves, control point and weights.	10
2	Wireframe modelling: Introduction to transformation of geometric models, visual realism, surface modelling- Bezier and B-spline surfaces. solid modelling techniques: Boundary representation, constructive solid geometry, analytical solid modelling, design and engineering applications. Standards for computer graphics: Standards for exchange images, data exchange standards- DFX, IGES, STEP, digital imaging and communication in medicine, STL conversion for additive manufacturing, introduction to reverse Engineering.	10

3	Introduction to NC systems and CNC: Machine axis and coordinate system, CNC machine tools, principle of operation CNC. CNC Part programming: fundamentals, machine origin, program origin, part origin, setting up of origin- absolute and incremental programming, preparatory codes and miscellaneous codes, detailed manual part programming for turning & milling operations using G codes and M codes, canned cycles and subroutines, cutting cycles, loops, sub program and macros, tool radius compensation CNC programming with interactive graphics, introduction of CAM packages.	12
4	Engineering analysis of solid models: Use of CAE in product development, discretization methods- FEM, FDM and FVM, CAE tools, basic steps in CAE. Element shapes: Nodes, nodal unknowns, coordinates systems, global coordinates, local coordinates: natural coordinates, shape functions, convergence requirement of shape functions, derivation of the shape functions using polynomials, numerical enumeration using 2 noded spring and bar elements Computer aided design optimization: Introduction to topology optimization, role of AI and ML in design optimization, generative design -overview of popular CAE software used in industry .	12

Suggestion on Project Topics:

- Solid modelling of consumer products using commercial CAD software.
- Experiment on CAD Customization (E.g., modelling of a customizable phone case).
- Familiarise with the reverse engineering procedures to develop required CAD models.
- Obtain 3D printable STL files from the DICOM model.
- Develop a CAM program to manufacture a complex part using CNC machining (E.g., a turbine blade).
- Evaluate the structural and mechanical integrity using CAE software.
- Compare the thermal performance of two different heat sinks using CAE software.
- Weight reduction of an existing design of an engineering component.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 subdivisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concepts of CAD/CAM and geometric modelling in design and manufacturing.	K3
CO2	Identify various forms of CAD models for solving engineering problems.	K3
CO3	Utilize CNC programming and CAM concepts for industrial applications.	K3
CO4	Analyze various engineering models to optimize the design of systems.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	-	-	-	-	-	-	-	-	3
CO2	3	2	1	-	1	-	-	-	-	-	-	3
CO3	3	2	1	-	2	-	-	-	-	-	-	3
CO4	3	3	3	-	3	-	-	-	-	-	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CAD/CAM Computer Aided Design and Manufacturing	M Groover, E. Zimmer	Pearson	1st Ed., 2014
2	Text book of finite element analysis	P Seshu	PHI learning	1st Ed., 2003
3	Practical Finite element Analysis	Nithin S Gokhale	Finite to infinite publishers, India	1st Ed., 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basics of Computer Aided Geometric Design: An Algorithmic Approach	M Ganesh	Dreamtech Press	3rd Ed., 2019
2	Mathematical Elements of Computer Graphics	David F Rogers, Adams J H	McGraw Hill	15th Reprint, 2008
3	Principles of computer aided design and Manufacturing	Farid M Amirouche	Pearson	2nd Ed., 2004
4	Computer Control of Manufacturing Systems	Yoram Koren	McGraw Hill	1st Ed., 1983
5	A first course in the Finite Element Method	Logan D L	Thomson-Engineering	1st Ed., 2012
6	Introduction to finite element method	J N Reddy	Mc Graw Hill India	4th Ed., 1999
7	A Hands-on introduction to Topology Optimization	Amir, M Mirzendehdel, Krishnan Suresh	Infinite books	1st Ed., 2017
8	Design theory and methods using CAD/CAE	Kuang-Hua Chang	Academic Press Inc	1st Ed., 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/102/112102
2	https://archive.nptel.ac.in/courses/112/102/112102
3	https://archive.nptel.ac.in/courses/112/104/112104289/
4	https://archive.nptel.ac.in/courses/112/102/112102101/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

APPLIED THERMAL ENGINEERING

Course Code	PEPET521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students about the concepts of gas power cycles and prime movers.
2. To familiarize the principles and working of pumps, compressors and heat exchangers.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of gas power cycles: Otto, Diesel, Brayton, and Rankine cycles, thermodynamic analysis and efficiency calculations, regenerative, intercooling, and reheating in gas power cycles, applications in power generation and propulsion systems.	9
2	Classification of prime movers: Internal combustion engines, steam turbines, and gas turbines, working principles and thermodynamic cycles, performance analysis and efficiency calculations, applications and selection of prime movers.	9
3	Types of pumps: Centrifugal, reciprocating, and rotary pumps, performance characteristics and selection criteria, pump performance curves and system curves, cavitation and its prevention. Types of compressors: Reciprocating, rotary, and centrifugal compressors, thermodynamic analysis of compressors, performance characteristics and selection criteria, efficiency and work input of compressors.	9
4	Heat Exchangers: Types of heat exchangers- shell and tube, plate and finned tube heat exchangers, heat transfer principles and design equations, measurement of effectiveness-NTU method and LMTD method, fouling and maintenance of heat exchangers.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concepts of gas power cycles and their thermodynamic characteristics.	K3
CO2	Apply the working principles of prime movers and their applications.	K3
CO3	Identify the performance characteristics of pumps and compressors.	K3
CO4	Develop the parameters of different types of heat exchangers.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	-	-	-	-	-	-	-	2
CO2	3	2	2	2	-	-	-	-	-	-	-	2
CO3	3	3	3	2	-	-	-	-	-	-	-	2
CO4	3	3	3	3	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Thermodynamics: An Engineering Approach	Yunus A Çengel, Michael A Boles	McGraw-Hill Education	9th Ed., 2019
2	Power Plant Engineering	P K Nag	McGraw-Hill Education	4th Ed., 2017
3	Applied Thermodynamics	Onkar Singh	New Age International Private Limited	4th Ed., 2018
4	A Textbook of Power Plant Engineering	R K Rajput	Laxmi Publications	5th Ed., 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulic Machines	Subramanya	McGraw Hill Education	1st Ed., 2017
2	Gas Turbines	V Ganesan	McGraw Hill Education	3rd Ed., 2017
3	I C Engines	V Ganesan	McGraw Hill Education	4th Ed., 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107291/

SEMESTER S5

MICRO ELECTRO MECHANICAL SYSTEMS

Course Code	PEPET522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide a comprehensive understanding of the fundamental principles and intrinsic characteristics of MEMS.
2. To introduce students about materials, fabrication process, packaging and integration strategies of MEMS.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to MEMS and Fabrication Technologies: Introduction, significance and new trends in micro and nano-scale systems, intrinsic characteristics of MEMS, elements of MEMS- microsensors and microactuators. Microelectronics Fabrication Process: Basics of the microelectronics fabrication process, materials for MEMS- silicon, polymers and metals used in MEMS, additional materials- silicon compounds (SiO₂, SiC, Si₃N₄), polycrystalline silicon, gallium arsenide, quartz, piezoelectric crystals, conductive polymers. Packaging and Integration: Glass encapsulation, MEMS process integration strategies.	9
2	Advanced Fabrication Techniques: Surface micromachining: sacrificial layer processes, micro motors, bulk micromachining- micro needles, micro nozzles, etching techniques-dry etching, plasma etching, wet etching. High Aspect-Ratio Processes: LIGA process, Deep Reactive Ion Etching (DRIE), thin film deposition-chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD), evaporation and sputtering.	9

3	Mechanical Concepts in MEMS: Crystal planes and orientation, internal force analysis, mechanical properties of silicon and related thin films, flexural beam bending analysis, torsional deflections, spring constant and resonant frequency. Electrical Concepts in MEMS: Semiconductor materials, calculation of charge carrier concentration, conductivity and resistivity of semiconductors.	9
4	Sensing Techniques: Electrostatic sensors, thermal sensors, piezoresistive sensors, piezoelectric sensors. Actuation Techniques: Thermal forces, shape memory alloys, piezoelectric crystals, electrostatic forces (parallel plate, torsion bar, comb drive actuators), micromechanical motors and pumps. Case Studies of MEMS: MEMS inertial sensors in automobiles (airbag deployment, automobile navigation), MEMS vibratory gyroscopes, MEMS accelerometers, MEMS devices in commercial applications.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental principles, characteristics of MEMS in various engineering and technology fields.	K2
CO2	Demonstrate various MEMS fabrication techniques.	K2
CO3	Explain mechanical and electrical concepts specific to MEMS.	K2
CO4	Apply various MEMS sensors and actuators in real world scenarios.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	-	-	-	-	-	2
CO2	3	-	-	-	-	-	-	-	-	-	-	2
CO3	3	3	-	-	-	-	-	-	-	-	-	2
CO4	3	3	2	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Foundation of MEMS	Chang Liu	Pearson Education	2nd Ed., 2011
2	MEMS and Microsystems Design and Manufacture	Tai Ran Hsu	McGraw Hill Education	1st Ed., 2017
3	Microsensors Mems and Smart Devices	Julian W. Gardner, Vijay K Varadan	Wiley India Pvt Ltd	1st Ed., 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	MEMS Materials and Processes Handbook	Reza Ghodssi, Pinyen Lin	Springer	1st Ed., 2011
2	Handbook of Silicon Based MEMS Materials and Technologies	Markku Tilli, Mervi Paulasto-Kröckel, Matthias Petzold, Horst Theuss, Teruaki Motooka, Veikko Lindroos	Elsevier	3rd Ed., 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/117/105/117105082/
	https://archive.nptel.ac.in/courses/108/106/108106165/

SEMESTER S5

DATA STRUCTURE ALGORITHMS

Course Code	PEPET523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the fundamental concepts of data structures and algorithms.
2. To enable the students to analyse basic algorithms for problem-solving.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic algorithm analysis: Time and space complexity, big O notation, arrays- definition, operations and applications, linked lists- singly and doubly linked lists, basic operations on linked lists, insertion, deletion, traversal.	9
2	Stacks- definition, operations, and applications, queues- definition, operations, and applications, implementation of stacks and queues using arrays and linked lists. Basic tree concepts- definitions, terminologies, and properties, Binary trees- representation and basic operations, Binary Search Trees (BST)- insertion, deletion, and searching.	9
3	Basic graph concepts: Definitions, terminologies, and representations, graph traversal algorithms- Depth-First Search (DFS) and Breadth-First Search (BFS). Shortest path algorithms- Dijkstra's and Floyd-Warshall algorithms. minimum spanning tree algorithms- Prim's and Kruskal's algorithms.	9
4	Sorting algorithms: Bubble sort, selection sort, insertion sort, searching algorithms- linear search, binary search, analysis and comparison of sorting and searching algorithms. Introduction to dynamic programming and greedy algorithms, simple examples and applications.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of data structures.	K2
CO2	Identify the algorithms applicable to the given context.	K3
CO3	Apply fundamental data structures and algorithms to solve simple engineering problems.	K3
CO4	Identify the relevant algorithms for sorting and searching applications.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	-	3	-	-	-	-	-	-	2
CO2	3	3	2	-	3	-	-	-	-	-	-	2
CO3	3	3	3	-	3	-	-	-	-	-	-	2
CO4	3	3	3	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Algorithm Design Manual	Steven S Skiena	Springer	2nd Ed., 2008
2	Introduction to Algorithms	Thomas H Cormen, Charles E Leiserson, Ronald L Rivest, Clifford Stein	MIT Press	4th Ed., 2022
3	Algorithms	Robert Sedgewick and Kevin Wayne	Addison Wesley	4th Ed., 2009
4	Data Structures and Algorithms in Python	Roberto Tamassia, Michael H Goldwasser, Michael T Goodrich	Wiley	1st Ed., 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Algorithms to Live By	Tom Griffiths	Henry Holt and Co	1st Ed., 2016
2	Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems	Martin Kleppmann	Shroff/O'Reilly	1st Ed., 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/102/106102064/

SEMESTER S5

LOGISTICS & SUPPLY CHAIN ENGINEERING

Course Code	PEPET524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the students to identify problems and solutions in supply chains.
2. To make the students find and solve problems related to logistics management.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Supply Chain: Objectives, structures, decision phases, performance measures, strategic fit, drivers and metrics. Design of distribution networks and models for facility location and layout decisions. Major features of postponement, mass customisation, supply chain restructuring, lean, agile, leagile, resilient, dual channel and international supply chains. Forecasting and aggregate planning models for supply chains.	9
2	Inventory planning in supply chain: Cycle inventory models, discounting models, multi-item inventory models and production-consumption models related to supply chains including multi-echelon systems. Safety inventory models for continuous review and periodic review cases of single and multi-items with probabilistic demand including for multi echelon supply chains. Models, tools and techniques related to optimal level of product availability, bullwhip effect and risk pooling.	9
3	Logistics management: Design options in logistics/transportation network, trade-offs, risk management, 3PL, 4PL and 5PL. Methods and techniques for allocation, routing, scheduling and sequencing in transportation. Formulation and solution of vehicle routing problems with quantitative examples.	9

4	Advanced logistics models: Mathematical formulation and solution of multi-stage transportation problems included with node capacities. Advanced logistics decision models including bin packing problems, fixed charge problems and knapsack problems. Methods, models and tools related to reverse, closed loop and green logistics in uncertain situations like pandemic and disaster scenarios.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the general features and design of supply chain networks.	K2
CO2	Apply techniques and models on inventory decisions in supply chains.	K3
CO3	Understand tools and models on logistics network decisions.	K2
CO4	Apply techniques and models on different logistics scenarios.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1	-	-	-	-	-	-	-	-	2
CO2	2	3	3	-	-	-	-	-	-	-	-	2
CO3	2	1	1	-	-	-	-	-	-	-	-	2
CO4	2	3	3	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Supply Chain Management Strategy, Planning and Operation	Chopra S, Meindl P	Pearson Education	6th Ed., 2016
2	Quantitative Models in Operations and Supply Chain Management	Srinivasan G	Prentice Hall India	2nd Ed., 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Logistical Management: The Integrated Supply Chain Process	Bowersox D J, Closs D J	McGraw Hill	1st Ed., 2017
2	Logistics and supply chain management,	Christopher M	Pearson Education	4th Ed., 2011
3	Supply Chain Management	Shah J	Pearson Education	2nd Ed., 2016
4	Manufacturing Operations and Supply Chain Management	Taylor D, Brunt D	Vikas Thomson Learning	1st Ed., 2009

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/106/110106045/
	https://archive.nptel.ac.in/courses/110/105/110105141/

SEMESTER S5

DESIGN OF EXPERIMENTS AND STATISTICAL ANALYSIS

Course Code	PEPET525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objective:

1. To familiarize with experimental design techniques that are applied to the data analysis process.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to design of experiments: Experimentation strategy, basic principles and application, simple comparative experiments, experiments with single factor analysis of variance- adequacy checking and interpretation of results, determination of sample size, random effect model, regression approach to analysis of variance.	9
2	Factorial experimentation: Factorial design, two factor factorial design- model adequacy checking, model parameters, sample size estimation, no interaction assumptions, general factorial design. 2^2 factorial design, 2^3 factorial design, introduction to 2^K design.	9
3	Fitting regression models: Linear regression, estimation of parameters, multiple regression, hypothesis testing, confidence intervals, random effect model introduction, response surface method, method of steepest ascent, analysis of second order response surface, experimental design for first and second order models.	9
4	Experiments with random factors: Random effect models, two factor factorial with random factor, two factor mixed model, rules for expected mean square, approximate F-Test, nested design, statistical analysis, diagnostic checking, variance components, staggered nested designs, design with both nested and factorial factors.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Examination	Analyse	Evaluate	Total
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> <i>a. Clearly defines the real-world quality issue.</i> <i>b. Examine and identify relevant contextual factors.</i>	5
	<i>Problem Analysis</i> <i>a. Present a structured realistic solution methodology.</i> <i>b. Compare and justify the proposed solutions with evidence and logical reasoning</i>	5
Evaluate	<i>Validation of Results</i> <i>a. Thoroughly evaluate the proposed solutions.</i> <i>b. Compares trade-offs, advantages, and disadvantages.</i> <i>c. Considers feasibility, scalability, and practical implications.</i>	5
	<i>Conclusion and Report Writing</i> <i>a. Summarizes procedure, findings and insights, limitation, and scope for future work.</i> <i>b. Preparation of Report with all components of project report.</i>	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the role of statistical methods in design and analysis of experiments.	K2
CO2	Apply the factorial design concepts to solve engineering problems.	K3
CO3	Develop various models based on the experimental data related to the given problems.	K3
CO4	Analyze the experimental data based on relevant parameters.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	-	-	-	-	-	-	-	3
CO2	3	2	1	2	-	-	-	-	-	-	-	3
CO3	3	2	-	3	2	-	-	-	-	-	-	3
CO4	3	2	1	1	-	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design and Analysis of Experiments	Douglas C Montgomery	Wiley India Pvt Ltd	8th Ed., 2013
2	Applied Multivariate Statistical Analysis	R A Johnson D W, Wichersn	PHI Learning Ptl Ltd	6th Ed., 2020
3	Optimal Design of Experiments	Peter Goos, Bradley Jones	John Wiley & Sons	1st Ed., 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Experiments for Engineers and Scientist	Jiju Antony	Elsevier Publications	3rd Ed., 2023
2	Design of Experiments: Statistical Principles of Research Design and Analysis	R O Kuehl	Duxbury/Thomson Learning, 2000	2nd Ed., 2000

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105087/

SEMESTER S5

INDUSTRIAL INSPECTION AND QUALITY CONTROL LAB

Course Code	PCPEL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To train students for accurate measurement and analysis of various parameters manufactured components.
2. To provide students with hands-on experience in quality control processes and techniques.

Expt. No.	Experiments (Minimum 10 Experiments should be completed)
1	Measurement of bore diameter using 2 – ball method and 4-ball method
2	Measurement of gear tooth profile using gear tooth vernier
3	Measurement of straightness and flatness using Autocollimator
4	Measurement of angle using sine bar
5	Measurement of screw thread and cutting tool angles using profile projector and tool makers microscope
6	Measurement of dimensional and geometrical features using CMM
7	Measurement of surface parameters using Talysurf
8	Calibration of LVDT
9	Measurement of force using load cell
10	Construction and interpretation of P chart from measurements.
11	Construction and interpretation of nP chart from measurements.
12	Construction and interpretation of X and R charts from measurements
13	Construction and interpretation of C chart from measurements.

14	Construction and interpretation of OC Curve.
15	Experiment on acceptance sampling.
16	Experiments to measure process capability of a process
17	Experiments on work study and standard time calculation.

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the competence in linear, angular and force measurements.	K3
CO2	Identify the gear and thread parameters of industrial products.	K3
CO3	Experiment with surface roughness and geometric parameters.	K3
CO4	Construct and analyse various quality control charts.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	-	-	-	-	-	2	-	2
CO2	3	2	2	2	-	-	-	-	-	2	-	2
CO3	3	2	2	2	-	-	-	-	-	2	-	2
CO4	3	2	2	2	-	-	-	-	-	2	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Metrology and Measurements	N V Raghavendra, L Krishnamurthy	Oxford University Press	1st Ed., 2013.
2	Introduction to Statistical Quality Control	Douglas C Montgomery	John Wiley & Sons	7th Ed., 2012.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Physics of Metrology	Alex Hebra	Springer Vienna	1st Ed., 2010
2	Mechanical Measurements	S P Venkateshan	Springer International Publishing	2nd Ed., 2021
3	A Textbook Of Engineering Metrology	I C Gupta	Dhanpat Rai Publications.	8th Ed., 2018
4	Engineering Metrology	R K Jain	Khanna Publishers	22nd Ed., 2022
5	Total Quality Management	P N Mukherjee	Prentice Hall India	1st Ed., 2006
6	Total Quality Management (TQM) - Principles, Methods, and Applications	Sunil Luthra, Dixit Garg, Ashish Agarwal, Sachin K Mangla	CRC Press	1st Ed., 2020
7	Statistical Quality Control	Mahajan M	Dhanpat Rai & Co.	2nd Ed., 2016
8	Quality Control and Improvement	A L Grant and R S Leavenworth	McGraw-Hill	7th Ed., 1996
9	Introduction to Industrial Engineering	Avraham Shtub, Yuval Cohen	Taylor and Francis	2nd Ed., 2017

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

PUMPS AND PRIME MOVERS LAB

Course Code	PCPEL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To provide students with hands-on training in the working of motors, generators and micro controllers.
2. To equip students with practical knowledge in operating different types of pumps, turbines and IC engines.

Expt. No.	Experiments
PART-A (Minimum 5 Experiments should be completed)	
1	Perform characteristic tests on induction motors.
2	Perform characteristic tests on DC motors.
3	Perform characteristic tests on DC generators.
4	Explain the speed control of the DC motor.
5	Develop programs to interface stepper motors with 8051 microcontrollers.
6	Develop assembly language programs for simple tasks using 8085 microprocessor kit.
PART-B (Minimum 5 Experiments should be completed)	
1	Performance testing of Pelton turbine.
2	Performance testing of Francis /Kaplan turbines.
3	Determination of performance characteristics of single and multi-stage centrifugal pumps.
4	Determination of performance characteristics of reciprocating pumps.
5	Performance analysis of a four-stroke diesel engine.
6	Performance analysis of a two-stroke engine.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva Voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Identify the operation characteristics and to estimate the efficiency of various electrical machines	K3
CO2	Apply the interfacing techniques and control algorithms for stepper motors in embedded systems.	K3
CO3	Explain the working principles of different types of pumps and prime movers.	K2
CO4	Analyze performance tests on pumps and prime movers and analyze the results.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	2	-	-	-	-	3	1	-	3
CO2	3	2	-	2	-	-	-	-	3	1	-	3
CO3	3	2	-	3	-	-	-	-	3	1	-	3
CO4	3	3	-	2	-	-	-	-	3	1	-	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Drives: Concepts and Applications	Vedam Subrahmanyam	McGraw Hill Education	2nd Ed., 2017
2	Power Plant Engineering	P K Nag	McGraw Hill Education	4th Ed., 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Motor Drives: Modeling, Analysis, and Control	R Krishnan	Pearson Education India	1st Ed., 2015
2	Internal Combustion Engines	V Ganesan	McGraw Hill Education	4th Ed., 2017

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

PRODUCTION ENGINEERING

SEMESTER: S6

OPERATIONS RESEARCH AND PRODUCTION MANAGEMENT

Course Code	PCPET601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Mins.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to use quantitative methods and techniques for effective decision making.
2. To familiarise the students with production/operations management.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of operations research(OR): Operation research models, solving the OR model, queuing and simulation models, art of modelling, phases of OR study. Modelling with linear programming(LP): two variable LP models, graphical LP solution, linear programming applications. Simplex method and sensitivity analysis: LP model, simplex method, Artificial starting solution, special cases in simplex method, sensitivity analysis, Duality and post-optimal analysis	11
2	Transportation Model: Definition, non-traditional transportation models, transportation algorithm, assignment model - Hungarian method. Queuing theory: Markovian queue models, single server & multiple server models. Game theory: Practical application of game theory, two-person, zero - sum games, solution with and without saddle point, rules of dominance, other solution methods.	11
3	Concepts of production and operations management (POM): Introduction, objectives, functions and scope of POM, types of production systems, historical evolution of POM, productivity, factors affecting productivity,	11

	recent trends in POM, competitiveness and operations strategies. Demand forecasting: Types of forecasting, demand patterns, forecasting models including moving average(simple and weighted), single exponential smoothing, double exponential smoothing, and regression techniques, measures of forecast accuracy.	
4	Aggregate planning and master production scheduling: Basic strategies for meeting uneven demand, techniques for aggregate planning, master scheduling. Inventory management: Functions of inventory, inventory problem classification, selective inventory control, continuous review and periodic review inventory control systems, deterministic models, quantity discount, economic order quantity. Material requirement planning (MRP): Concepts, MRP calculations, lot sizing in MRP, MRP-II, ERP, JIT and Lean Operations.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Solve decision problems using linear programming.	K3
CO2	Solve transportation, assignment and queuing models; solve problems using game theory.	K3
CO3	Explain the concepts of production and operations management; apply demand forecasting models to solve practical problems.	K3
CO4	Develop production plans and inventory policies.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	-	-	-	-	-	-	3
CO2	3	3	3	3	2	-	-	-	-	-	-	3
CO3	2	2	2	-	-	-	-	-	-	-	-	3
CO4	3	2	2	-	-	-	-	-	-	-	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operations Research: An Introduction	Hamdy A Taha	Pearson/Prentice Hall	10th Ed., 2019
2	Operations Management	William J Stevenson	McGraw Hill	14th Ed., 2022

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to operations search	Federick S Hiller , Gerald J Lieberman	Tata McGraw-hill	9th Ed., 2010
2	Operations research	P Rama Murthy	New Age	2nd Ed., 2007
3	Operations research	G. Srinivasan	PHI	2nd Ed., 2010
4	Production and operations management	R. B. Khanna	PHI	2nd Ed., 2015
5	Production and operations management	R. Panneerselvam	PHI	3rd Ed., 2012
6	Operations management	Norman Gaither and Greg Frazier	Cengage Learning	9th Ed., 2004
7	Production and operations management	Everett E. Adam, Jr. and Ronald J. Ebert	PHI	5th Ed., 2009
8	Modern production\ operations management	Elwood S. Buffa and Rakesh K, Sarin	John Wiley and Sons	8th Ed., 2007

Video Links (NPTEL, SWAYAM...)	
Link ID	https://nptel.ac.in/courses/110106062
	https://archive.nptel.ac.in/courses/110/107/110107141/

SEMESTER: S6

METAL FORMING AND TOOL DESIGN

Course Code	PCPET602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Mins.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the principles and mechanics of metal forming processes.
2. To enable the students to design tools and dies for different engineering applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Characteristics of metal forming and applications: Classification of metal forming processes according to stresses, theory of plastic deformation, strain hardening, effect of temperature, composition and strain rates on metal forming, recovery, recrystallization and grain growth, characteristics and applications of hot working and cold working. Compressive forming: Working principle, characteristic features, capabilities and application of processes like, rolling, forging and extrusion. Combined tensile and compressive forming: Working principle, characteristic features, capabilities and application of processes like, wire drawing, tube drawing, deep drawing or cupping and redrawing, spinning, friction and lubrication in metal forming. Common defects in metal forming, Erichsen cupping test.	9
2	Forming by bending: Working principle, characteristic features, capabilities and applications-V-bend, U-bend, roll bending, straightening, bend allowance and bend length, spring back in bending and methods to counter spring back, defects in bending and remedies. Forming by shearing: Working principle, characteristic features, capabilities and applications - blanking, punching, notching, mechanism of shearing, punch & die design for blanking and punching, operations, estimation of	9

	shearing load and press capacity, efficient utilization of sheet. High energy rate forming processes: Working principle, characteristic features, process variables, capabilities and applications of processes like explosive forming, hydroforming, magneto forming etc.	
3	Jigs, fixtures and gauge design: Limits, fits and tolerances, limit gauge design, geometric tolerances, methods for verification of straightness, and flatness in machine tools, use of interferometry in straightness, and flatness measurements. Different types of locating and clamping devices: Basic principles of location and design, locating methods and devices, basic principles of clamping, classification of jigs and fixtures and their applications.	9
4	Design of plastic moulding dies: Plastic materials, shrinkage, two and three plate mould design, standard mould plates, parting line, core and cavity generation, runner and gate design, mould cooling, ejection methods, tool materials, runner less moulds. Design of blow moulding and extrusion dies. Design of forging and casting dies: Design considerations for forging, pressure die casting, powder metallurgy and rubber moulding.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the metal forming concepts and identify the parameters influencing metal forming.	K3
CO2	Identify the principles of the sheet metal working.	K3
CO3	Make use of the principles of designing Jigs, fixtures and gauges.	K3
CO4	Develop dies for a given engineering application.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	-	2	-	-	-	-	-	3
CO2	3	-	2	-	-	2	-	-	-	-	-	3
CO3	3	-	2	-	-	2	-	-	-	-	-	3
CO4	3	-	2	-	-	2	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Technology: Foundry, Forming and Welding	Rao P N	Tata McGraw Hill	5th Ed., 2018
2	Tool Design	V C Goold, George H LeCain, Cyril Donaldson	McGraw Hill Education	5th Ed., 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Processes	Kaushish J P	PHI Learning Private Ltd.	2nd Ed., 2010
2	Manufacturing Science	Ghosh, Amitabh & Mullick	EWP	2nd Ed., 2010
3	Manufacturing Engineering & Technology	Kalpkjian S & Schmid Stevens	Pearson	7th Ed., 2013
4	DeGarmo's Materials and Processes in Manufacturing	Black J T & Kohser Ronald A	John Wiley & Sons, Inc.	10th Ed., 2007
5	Fundamentals of Metal Forming	Juneja B L	New Age International (P) Ltd	1st Ed., 2007
6	A Textbook of Production Engineering	P C Sharma	S Chand	11th Ed., 2009
7	Tool Engineering and Design	G R Nagpal	Khanna Publishers	1st Ed., 2000
8	Machine Tools Handbook: Design and Operation	Prakash Hiralal Joshi	McGraw Hill Education (India) Private Limited	1st Ed., 2007
9	Injection Mould Design	Pye R G W	East-West Press Pvt.	1st Ed., 2000
10	Fundamentals of Tool Design	Frank W Wilson	Literary Licensing, LLC	1st Ed., 2012
11	Fundamentals of Tool Design	John G Nee, William Dufraine, John W Evans, Mark Hill	Society of Manufacturing Engineers	6th Ed., 2010

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107250/
	https://archive.nptel.ac.in/courses/107/103/107103012/

SEMESTER: S6

REFRIGERATION AND AIR CONDITIONING

Course Code	PEPET631	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of the principles and applications of refrigeration and air conditioning systems.
2. To familiarise fundamental concepts of refrigeration systems, psychrometry, and air conditioning system design.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Refrigeration: History, applications, and importance of refrigeration. Thermodynamics of Refrigeration: Reversed Carnot cycle, heat pumps, limitations of the reversed Carnot cycle. Units of Refrigeration: Definition and practical examples. Air Refrigeration Systems: Reversed Joule cycle, aircraft refrigeration systems (simple, bootstrap and regenerative systems). Special Refrigeration Systems: Thermoelectric Refrigeration (Peltier Effect), Magnetic Refrigeration.	9
2	Vapour Compression Systems: Simple cycle, representation on T-s and P-h diagrams, Coefficient of Performance (COP), and methods to improve COP (superheating, undercooling, liquid suction heat exchanger), cascade systems. Vapour Absorption Systems: Ammonia-water system, Lithium bromide-water systems, Electrolux systems, comparison with vapour compression systems and drawbacks.	9

	Refrigerants: Properties, eco-friendly refrigerants, mixed refrigerants, selection criteria for different applications, refrigerant leakage detection, refrigerant charging.	
3	Psychrometry: Properties of air (saturated, unsaturated, dry, wet, dew point, humidity levels), thermodynamic equations, adiabatic saturation process, psychrometric charts, processes (sensible heating/cooling, humidifying/dehumidifying), air washers, bypass factors, sensible heat factor, RSHF, GSHF. Introduction to ventilation: Fundamentals of good indoor air quality Need for ventilation, Effects of Relative humidity in ventilation, Control of microbial growth. Types of ventilation system. Supply system - Devices used in supply systems, Air Inlet system. Filters heating & cooling equipment, Fans, Duct, Grills, Diffusers Exhaust system General exhaust systems. Removal of pollutants and contaminated air. Air cleaning devices.	9
4	Air Conditioning Systems: Comfort and Industrial air conditioning. Comfort air conditioning factors, effective temperature, comfort charts, cooling load estimation, design conditions, supply conditions, fresh air supply, thermodynamics of the human body. Applications of Air Conditioning: Room air conditioners, split systems, packaged systems, all-air systems, chilled water systems, winter and summer air conditioning, heating systems, humidifiers, year-round AC system and controls (thermostat, humidistat), air distribution systems (duct design), air conditioning for various environments (restaurants, hospitals, retail outlets, computer centres, theatres, industrial applications).	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the basic principles of refrigeration systems and identify their practical applications.	K3
CO2	Make use of vapour compression and vapour absorption refrigeration systems with suitable refrigerants for various applications.	K3
CO3	Explain the concepts of psychrometric processes and illustrate the principles of ventilation systems.	K2
CO4	Make use of the principles of air conditioning systems for various applications.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	1	2	2	-	-	-	-	1
CO2	3	2	2	-	1	2	2	-	-	-	-	1
CO3	3	-	-	-	1	2	2	-	-	-	-	1
CO4	3	-	2	-	1	2	2	-	-	-	-	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Refrigeration and Air Conditioning Technology	Bill Whitman, Bill Johnson, John Tomczyk	Cengage Learning	8th Edition, 2023
2	Refrigeration and Air Conditioning	G.F. Hundy, A.R. Trott, T.C. Welch	Butterworth-Heinemann	5th Edition, 2021
3	Refrigeration and Air Conditioning	C.P. Arora	McGraw Hill Education	5th Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Refrigeration and Air Conditioning	Andrew D. Althouse, Carl H. Turnquist, Alfred F. Bracciano	Goodheart-Willcox	21st Edition, 2022
2	Air Conditioning Engineering	W.P. Jones	Arnold	5th Edition, 2013
3	Refrigeration and Air Conditioning Technology Handbook	J. Whitfield, P. Dincer	CRC Press	2nd Edition, 2020
4	Handbook of Air Conditioning and Refrigeration	Shan K. Wang	McGraw-Hill	3rd Edition, 2017
5	Practical Guide to Refrigeration and Air Conditioning	Minnick & Associates	Delmar Cengage Learning	2nd Edition, 2016
6	Fundamentals of HVACR	Carter Stanfield, David Skaves	Pearson	3rd Edition, 2016
7	Air Conditioning and Refrigeration Repair	Roger A. Fischer, Ken Chernoff	McGraw-Hill	2nd Edition, 2014
8	Textbook of Refrigeration and Air Conditioning	R.K. Rajput	S. Chand Publishing	6th Edition, 2016
9	Principles of Refrigeration	Roy J. Dossat	Pearson Education	5th Edition, 2019
10	Air Conditioning Principles and Systems	Edward G. Pita	Prentice Hall	4th Edition, 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107208/
	https://nptel.ac.in/courses/112105129

SEMESTER: S6

FINITE ELEMENT ANALYSIS

Course Code	PEPET632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the students to acquire the basic concepts of finite element formulation methods.
2. To familiarise finite element methodologies through simple structural and heat transfer problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to finite element method (FEM): FEM as a numerical tool for design, basic concepts, formulation procedures. FE modelling direct approach: One dimensional bar element, element stiffness, assembly of elements, properties of stiffness matrix, treatment of boundary conditions, temperature effects, stress computation, support reaction analogous one dimensional problems of torsion and heat conduction. Beam element: Beam relationships, one dimensional beam element FE formulation, element stiffness matrix, load considerations, boundary conditions, member end forces.	9
2	FE modelling Direct approach: Plane truss element formulation, coordinate transformation, local and global coordinates, element matrices, assembly of elements, treatment of boundary conditions, stress calculation, simple problems. Convergence requirements, patch test. Variational methods: Functionals, weak and strong form, essential and non-essential boundary conditions, principle of stationary potential energy, Rayleigh-Ritz method, simple examples.	9
3	Interpolation: Shape function, Lagrange interpolation. FE formulation from virtual work principle: Displacement differentiation matrix [B-matrix], element matrices for bar and constant strain triangle	9

	element [CST element], load considerations, consistent nodal loads, simple problems. Two dimensional FE formulation: Two dimensional steady state heat conduction, element matrices for a triangular element, boundary conditions, simple problems. FE formulation for two dimensional stress analysis from potential energy, element matrices, plane bilinear element.	
4	Weighted residual methods: Galerkin FE formulation, axially loaded bar – heat flow in a bar. Isoparametric formulation: Natural coordinates, linear and quadratic bar element, linear triangle and plane bilinear elements for scalar fields, jacobian matrix, element matrices, Gauss quadrature, requirements for isoparametric elements, accuracy and mesh distortion.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Utilise finite element method for one-dimensional problem formulation and develop solution.	K3
CO2	Apply the finite element formulation using the truss element method and variational methods.	K3
CO3	Utilise finite element method for two-dimensional problem formulation and develop solution.	K3
CO4	Apply weighted residual method and isoparametric formulation in finite element analysis.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3	-	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	-	-	-	-	2
CO3	3	3	2	2	3	-	-	-	-	-	-	2
CO4	3	3	2	2	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text book of finite element analysis	P Seshu	PHI Learning Private Limited New Delhi	1st Ed., 2012
2	Finite element analysis	S. Senthil	Lakshmi publications Chennai	3rd Ed., 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Finite Element Analysis for Engineering and Technology	T. R. Chandrupatla	University Press	2nd Ed., 2004
2	Fundamentals of Finite Element Analysis	D. V. Hutton	Tata McGraw Hill	1st Ed., 2005
3	Finite Element Analysis	S. S. Bhavikatti	New Age International	4th Ed., 2021

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/104/112104193/
	https://archive.nptel.ac.in/courses/112/104/112104205/

SEMESTER: S6

DEEP LEARNING

Course Code	PEPET633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PEPET413: Machine learning	Course Type	Theory

Course Objectives:

1. To familiarise with deep learning frameworks and tools.
2. To enable students to implement various deep learning architectures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of deep learning and its applications: Differences between machine learning and deep learning, comparisons in applications. Introduction to neural networks: Neurons, layers, activation functions, concepts of forward and backward propagation.	8
2	Overview of popular deep learning frameworks: TensorFlow, Keras, PyTorch. Setting up the environment and basic operations. Building and training a simple neural network using TensorFlow/Keras/PyTorch. Introduction to deep feedforward networks: Convolutional neural networks (CNNs) - architecture, operations, and applications, recurrent neural networks (RNNs) - architecture, operations, and applications.	10
3	Data preprocessing and augmentation: Loss functions and optimization algorithms, techniques to avoid overfitting - Regularization, dropout, and batch normalization, identification of suitable algorithms.	9
4	Transfer learning and pre-trained models: Generative adversarial networks (GANs), concepts and applications, autoencoders and their applications in anomaly detection and compression. Case studies of deep learning applications in various industries.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the principles and concepts of deep learning.	K2
CO2	Demonstrate the basic and advanced deep learning architectures.	K2
CO3	Identify deep learning algorithms for various applications.	K3
CO4	Utilise the deep learning frameworks and tools for developing and testing models.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	1	2	-	-	-	-	2
CO2	3	-	-	-	-	1	2	-	-	-	-	2
CO3	3	-	3	-	-	2	2	-	-	-	-	3
CO4	3	-	3	-	-	2	2	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep Learning	Ian Goodfellow, Yoshua Bengio	MIT Press	1st Ed., 2016
2	Deep Learning with Python	Francois Chollet	Manning Publications	2nd Ed., 2021
3	Deep Learning from Scratch: Building with Python from First Principles	Seth Weidman	Shroff/O'Reilly	1st Ed., 2019
4	Deep Learning for Coders with Fastai and PyTorch: AI Applications Without a PhD	Jeremy Howard, Sylvain Gugger	Shroff/O'Reilly	1st Ed., 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Generative Deep Learning: Teaching Machines To Paint, Write, Compose, and Play	David Foster, Karl Friston	Shroff/O'Reilly	2nd Ed., 2019
2	Designing Machine Learning Systems: An Iterative Process I for Production-Ready Applications	Chip Huyen	Shroff/O'Reilly	1st Ed., 2022

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/106/106106184/
	https://archive.nptel.ac.in/courses/106/106/106106201/

SEMESTER: S6

PLANT LAYOUT AND DESIGN

Course Code	PEPET634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart knowledge about the facilities planning process, layout and layout design.
2. To introduce the concepts of material handling systems, plant safety and equipment replacement.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to facilities planning: Objectives, selection of site, plant layout - principles and objectives of good layout, types of layouts - product, process, fixed position, combination and cellular layouts. Layout Design: Major considerations, plant layout procedure, layout design of factories, office, storage area etc., consideration of working people, storage facilities and general equipment for amenities.	9
2	Systematic layout planning (SLP): Procedure, from-to chart, activity relationship chart, space relationship diagram. Design of assembly lines, line balancing, simple problems, computer applications in layout designs. Environmental aspects of design: Environmental factors like lighting, ventilation, dust control and humidity.	9
3	Plant safety: Elements of industrial safety, causes and prevention of accidents, pollution and environmental considerations, considerations for different types of plant services like steam, compressed air etc. Material handling system and equipment: Material handling in plants, principles of material handling, activities and functions, unit load concept. stores and warehouses, receiving and dispatching areas, choice of material handling equipment.	10

4	Material handling cost and economical aspects: Depreciation- methods, simple problems. Equipment replacement: Repair, replacement based on technical and economical consideration, methods of replacement studies, simple problems, break-even analysis.	8
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concepts of facilities planning and layout design.	K2
CO2	Apply the lay out design concepts in industries and explain plant services and environmental aspects of plant design.	K3
CO3	Demonstrate the material handling systems and explain the importance of safety in industries.	K2
CO4	Apply the economical aspects of material handling and equipment replacement.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	2	-	-	-	-	2
CO2	3	-	2	-	1	-	2	-	-	-	-	2
CO3	3	-	2	-	-	-	2	-	-	-	-	2
CO4	3	-	2	-	-	-	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Plant Layout and Material Handling	James M Apple	Krieger Pub Co.	2nd Ed., 2011
2	Plant Layout and Materials Handling	Pemberton A W	John Wiley	1st Ed., 2017
3	Plant Layout and Materials Handling	S C Sharma	Khanna publishers	1st Ed., 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Facilities Planning and Design	Alberto Garcia-Diaz, James MacGregor Smith	Pearson Education Limited	1st Ed., 2014
2.	Plant Layout and Material Handling	Fred E Meyers	Prentice Hall	1st Ed., 1993

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107292/
	https://archive.nptel.ac.in/courses/103/105/103105166/

SEMESTER: S6

FINANCIAL ENGINEERING

Course Code	PEPET635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course type	Theory

Course Objectives:

1. To provide students with knowledge of financial instruments and their application.
2. To equip students with the skills to analyse and manage financial risks.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Financial markets and instruments: Time value of money and interest rates, introduction to financial statements and analysis, introduction to financial accounting principles. Overview of Indian accounting standards: Preparation of financial statements, balance sheet, profit and loss account and cash flow statement.	9
2	Fixed Income Securities: Bond pricing and yields, term structure of interest rates, duration and convexity, introduction to mortgage-backed securities. Risk management tools and techniques: Types of financial risks - market, credit, operational and liquidity risks, risk - variance & covariance matrix, correlation matrix & portfolio variance, value at risk (VaR) and its computation.	9
3	Derivatives: Futures and options, pricing of forward and futures contracts, introduction to option pricing - Binomial model, Black-Scholes model. Option Greeks: Volatility smiles and volatility surfaces, hedging strategies using derivatives, applications in Indian scenario.	9
4	Valuation and valuation techniques: Fundamental principle of value creation, risk and cost of capital, ROIC, growth, valuation techniques - enterprise discounted cash flow model, economic profit-based valuation	9

	models, adjusted-present-value model, capital cash flow model, cash-flow-to-equity valuation model. Implementation of financial models using computational tools (Python, R).	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> <i>a. Clearly defines the real-world quality issue.</i> <i>b. Examine and identify relevant contextual factors.</i>	5
	<i>Problem Analysis</i> <i>a. Present a structured realistic solution methodology.</i> <i>b. Compare and justify the proposed solutions with evidence and logical reasoning</i>	5
Evaluate	<i>Validation of Results</i> <i>a. Thoroughly evaluate the proposed solutions.</i> <i>b. Compares trade-offs, advantages, and disadvantages.</i> <i>c. Considers feasibility, scalability, and practical implications</i>	5
	<i>Conclusion and Report Writing</i> <i>a. Summarizes procedure, findings and insights, limitation and scope for future work.</i> <i>b. Preparation of report with all components of project report.</i>	5

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 subdivisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of financial markets, instruments, and their applications	K2
CO2	Assess and mitigate financial risks through various hedging strategies and risk management tools.	K3
CO3	Apply derivative hedging strategies for optimizing investments and resources.	K3
CO4	Apply valuation and valuation techniques; evaluate the financial models developed with computational software.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	-	-	2	-	2	2
CO2	3	-	2	-	-	-	-	-	2	-	2	2
CO3	3	-	3	-	2	-	-	-	3	-	2	2
CO4	3	-	3	-	3	-	-	-	3	-	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Options, Futures & Other Derivatives	John C. Hull, Sankarshan Basu	Pearson Education	10th Ed., 2018
2	Valuation: Measuring and Managing the Value of Companies	McKinsey & Company Inc., Tim Koller, Marc Goedhart	Wiley	7th Ed., 2020
3	Option Volatility and Pricing: Advanced Trading Strategies	Sheldon Natenberg	Fragile Books	2nd Ed., 2014
4	Investment Valuation: Tools and Techniques for determining The Value Of any asset	Aswath Damodaran	Wiley	3rd Ed., 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Corporate Finance	Richard A. Brealey	McGraw Hill	1st Ed., 2023
2	Romancing The Balance Sheet	Dr Anil Lamba	Harper Business	2nd Ed., 2019
3	The Psychology of Money: Timeless Lessons on Wealth, Greed, and Happiness	Morgan Housel	Jaico Publishing House	1st Ed., 2020

Video Links	
Link ID	https://archive.nptel.ac.in/courses/110/104/110104066/

SEMESTER: S6**IoT AND AUTOMATION**

Course Code	PBPET604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	4	ESE Marks	40
Credits	3:0:0:1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the pneumatic and hydraulic automation systems, sensors and transducers.
2. To equip students to develop IoT systems and network models.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Pneumatic and hydraulic automation: Automation definition, types and strategies fluid power, hydraulics and pneumatic systems, basic elements, power pack, basic pneumatic systems. Rotary pumps and compressors: Gear, lobe and vane pumps, screw, lobe and vane compressors, direction control valves, actuators - types of cylinders and hydraulic motors, direction control valves, circuit representation using symbols - single and double cylinder circuits, servo control system.	12
2	Electric and electronic controls: Sensors and transducers - classification, thermocouple, thermistor, photo sensors, inductive sensors, pressure, strain and piezoelectric sensors. microprocessor, micro controllers, PLC controller, PLC Programming.	10
3	Internet of Things (IoT): Definition, architecture, applications, wireless networks, hardwares, event processing, organisation of IoT system, subscriber publish model, protocol, time base and database, device design space, comparison of cost of ownership and power with conventional systems.	11
4	Event driven system analysis: IoT network model, IoT event analysis, Industrial IoT (IIoT), industry 4.0, IIoT architecture, industrial internet consortium, functional domains, RFID, WSN technologies, IIoT	11

	applications, challenges. System security and application security, ARMET approach, examples.	
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Suggestion on Project Topics

- Selection of hydraulic cylinders for specific applications.
- Identification of robotised motion and mapping of the same into multiple cylinder actuation.
- Design of hydraulic or pneumatic circuits for actuation of multiple cylinders.
- Design of electro- pneumatic circuits for multiple cylinder actuation using ladder logic diagrams.
- PLC programming for any specific process.
- Design of IoT based systems for automation and comparing with conventional systems in cost terms.
- It enabled automation of machine operation leading to unmanned operations.

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 subdivisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop automated solution for operations using hydraulic and pneumatic actuators	K3
CO2	Develop automated solutions for moderately complex operations using PLC programming	K3
CO3	Apply IoT based systems for specific industrial operations	K3
CO4	Develop data capturing systems and implement industry 4.0 for shop floor operations	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	2	-	-	-	-	-	-	-	-	2
CO2	2	-	3	-	-	-	-	-	-	-	-	2
CO3	3	-	3	2	-	-	-	1	1	-	-	2
CO4	3	-	3	2	-	-	-	1	1	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Automation and Robotics	A K Gupta & S K Arora	Mercury Learning and Information	1st Ed., 2013
2	Internet-of-Things (IoT) Systems: Architectures, Algorithms, Methodologies	D S Mariyne Wolf	Springer	1st Ed., 2018
3	Automation, Production Systems, and Computer-Integrated Manufacturing	Mikell P Groover	Pearson	4th Ed., 2014
4	Internet of Things	Shriram K V, Abhishek S. Nagarajan & R. M. D. Sundaram	John Wiley Pubs	2nd Ed., 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotics automation and control in industrial service setting	Zong wei Luo	IGI Global	1st Ed., 2015
2	Designing the internet of things	Adrian, M. and Hakim C	John Wiley & Sons Inc	1st Ed., 2013

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/105/106105195/
	https://archive.nptel.ac.in/courses/106/105/106105166/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

DESIGN THINKING AND PRODUCT DEVELOPMENT

(Common to Group C & Group D)

Course Code	GZEST605	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	60
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To guide students through the iterative stages of design thinking, including empathizing with users, defining problems, ideating solutions and developing Proof of Concepts (PoC) and technical feasibility studies.
2. To promote the development of critical thinking skills by engaging students in integrative inquiry, where they ask meaningful questions that connect classroom knowledge with real-world applications.
3. To equip students with the ability to involve in product design considering the sustainability, inclusivity, diversity and equity aspects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of design thinking and product development: Overview of stages of product development lifecycle; Design thinking -Definition-Design thinking for product innovation; Bringing social impact in ideation- Identifying societal needs-understanding multi-faceted issues-community engagement and empathetic design- technological innovation meeting societal needs; Understanding and Bridging the divide using Human Centered Design (HCD); Designing for inclusivity in product development- embracing user diversity - Long term impact - sustainability encompassing environmental, economic and social dimensions; Technology Readiness Level in the Innovation Life-cycle; Performing a self-check on innovative ideas - Originality of idea- understanding innovation landscape -	6

	patentability - understanding the economic landscape - Unique Selling Proposition (USP) - Repeatability and Manufacturability - Sustainability - Leveraging business models for comprehensive analysis	
2	Empathize: Design thinking phases; Role of empathy in design thinking; Methods of empathize phase - Ask 5 Why/ 5 W+H questions; Empathy maps - Things to be done prior to empathy mapping - Activities during and after the session; Understanding empathy tools - Customer Journey Map - Personas. Define: Methods of Define Phase: Storytelling, Critical items diagrams, Define success.	6
3	Ideation : Stages of ideation; Techniques and tools - Divergent thinking tools - Convergent thinking tools - Idea capturing tools; Cross-industry inspiration; Role of research in ideation - Market research - consumer research - leveraging research for informed ideation; Technological trends - navigating the technological landscape - Integrating emerging technologies; Feasibility studies - technical, economic, market, operational, legal, and ethical feasibility; Ideation session- techniques and tips. Proof of Concept (PoC) : Setting objectives; Risk assessment; Technology scouting; Document and process management; Change management; Knowledge Capture; Validating PoC; Story telling in PoC presentation	6
4	Design: Navigating from PoC to detailed design; Developing Specification Requirement Document (SRD); Design for manufacturability; Design to cost; Pre-compliance; Design Failure Mode and Effects Analysis (DFMEA); Forecasting future design changes. Prototyping: Alpha prototypes; Beta prototypes; Transition from design to prototype; Goals and expectations for Alpha and Beta prototypes; Effective strategies for maintaining timeline in prototyping; Testing and refining Alpha prototypes; Transitioning to Beta prototypes. Pilot build: Definition and purpose of a pilot build; setting objectives; Identification and selection of manufacturing partner for pilot build; Testing procedures in pilot build; Scaling from pilot build to full-scale production / implementation.	6

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Empathize to capture the user needs and define the objectives with due consideration of various aspects including inclusivity, diversity and equity	K5
CO2	Ideate using divergent and convergent thinking to arrive at innovative ideas keeping in mind the sustainability, inclusivity, diversity and equity aspects.	K6
CO3	Engage in Human Centric Design of innovative products meeting the specifications	K5
CO4	Develop Proof of Concepts (PoC), prototypes & pilot build of products and test their performance with respect to the Specification Requirement Document.	K4
CO5	Reflect on professional and personal growth through the learnings in the course, identifying areas for further development	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		2	3	3	3	2	2		3
CO2	3	2	3		2	3	3	3	2	2		3
CO3	3	2	3		2	3	3	2	2	2		3
CO4	3	2	2		3	3	3	2	2	2		3
CO5	3					3	3	2	2	2		3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product Sense: Engineering your ideas into reality	Dr. K R Suresh Nair	NotionPress.com	2024
2	Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation	Tim Brown	HarperCollins Publishers Ltd.	2009
3	Design Thinking for Strategic Innovation	Idris Mootee	John Wiley & Sons Inc.	2013

Sample Assignments:

1. Evaluate and prepare a report on how the aspects including inclusivity, diversity and equity are taken into consideration during the empathize and define phases of the Miniproject course.
2. Evaluate and prepare a report on how the aspects including sustainability, inclusivity, diversity and equity are taken into consideration during the ideate phase of the Miniproject course.
3. Evaluate and prepare a report on how User-Centric Design (UCD) is used in the design and development of PoC of the product being developed in the Miniproject course.
4. Prepare a plan for the prototype building of the product being developed in the Miniproject course.
5. Report on the activities during the empathize phase including the maps & other materials created during the sessions.
6. Report on the activities during the define phase including the maps & other materials created during the sessions.

SEMESTER: S6

INDUSTRIAL QUALITY CONTROL

Course Code	OEPET611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of the fundamental principles and practices of industrial quality control.
2. To equip students with the skills to use advanced statistical methods for quality improvement and decision-making.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to industrial quality control: Quality basics, meaning of quality, factors affecting quality, quality principles, quality function, quality control: aims and objectives of quality control, cost of quality, seven QC tools, historical perspective of quality control.	9
2	Statistical quality control: Introduction, concept of variability, types of control charts, measurement of control limits, control charts for variables, large sample data, warning limits, revised control limits, group control chart, control chart with line trend.	9
3	Control charts for attributes: Control charts for non-conforming models, control charts for fraction non-conforming, process and measurement system. Capability analysis: Using probability plot, process capability ratios, specification limits and tolerance.	9
4	Acceptance sampling: Advantages and disadvantages of sampling methods, sampling techniques, sampling risks and indices, operating characteristics curves, average outgoing quality limits. Sampling plans: Simple, double, multiple and sequential sampling plans, tightened inspection, Dodge-Romig system.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the fundamental concepts and principles of quality and quality control.	K2
CO2	Apply statistical methods to construct and interpret control charts for variables.	K3
CO3	Construct and interpret control charts for attributes, and to identify process capability parameters.	K3
CO4	Utilize acceptance sampling methods and develop sampling plans to enhance quality control processes.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	1	-	-	-	-	-	2
CO2	3	2	-	-	-	1	-	-	-	-	-	2
CO3	3	2	-	-	-	1	-	-	-	-	-	2
CO4	3	2	-	-	-	1	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Statistical Quality Control	Douglas C Montgomery	John Wiley & Sons	7th Ed., 2012
2	Fundamentals of Quality Control and Improvement	Amitava Mitra	John Wiley & Sons	3rd Ed., 2008
3	Statistical Quality Control	M Mahajan	Dhanpat Rai & Co	1st Ed., 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Quality Control and Improvement	A L Grant and R S Leavenworth	McGraw-Hill	7th Ed., 1996
2	Statistical Process Control and Quality Improvement	Gerald M Smith	Pearson	5th Ed., 2004

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105088/

SEMESTER: S6

INDUSTRIAL ENGINEERING AND PRODUCTION MANAGEMENT

Course Code	OEPET612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise students with principles and concepts of Industrial Engineering.
2. To equip students with the concepts of production management and its applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to industrial engineering – Evolution of modern Concepts in Industrial Engineering, application of industrial engineering-functions and role of an industrial engineer. Productivity: Concept and definition, productivity and standard of living, techniques for productivity improvement, importance of work study towards productivity improvement, measuring productivity of an enterprise, fractional and multi factor productivity.	9
2	Introduction to plant layout: Definition, objective and importance. types of layout - product, process, fixed position and cellular layout, factors influencing plant layout - material flow, equipment, safety and ergonomics. Material handling: Principles, types of material handling equipment, selection and applications.	9
3	Introduction to production management: Definition, objectives and scope of production management, types of production systems - job, batch, mass, and continuous production, emerging trends in production management. Inventory management: Functions of inventory, inventory problem classification, selective inventory control, continuous review and periodic review inventory control systems, deterministic models, quantity discount,	9

	economic order quantity.	
4	Planning and control of production management: Demand forecasting - types of forecasting, demand patterns, forecasting models including moving average (simple and weighted), exponential smoothing and regression techniques, measures of forecast accuracy. Aggregate planning and master production scheduling: Basic strategies for meeting uneven demand, techniques for aggregate planning, master scheduling.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the basic concepts of industrial engineering and productivity.	K3
CO2	Demonstrate the types of plant layouts and the principles of material handling.	K2
CO3	Apply the concepts of production management and inventory models.	K3
CO4	Explain the concepts of production planning and control of operations.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	2	-	-	-	-	-	2
CO2	3	-	-	-	-	2	-	-	-	-	-	2
CO3	2	2	-	-	-	2	-	-	-	-	-	2
CO4	3	-	-	-	-	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering and management	Dr B Kumar	Khanna Publishers	10th Ed., 2004
2	Industrial Engineering and management	O P Khanna	Danpat Rai publication	2018
3	Operations management	Norman Gaither and Greg Frazier	Cengage Learning	9th Ed., 2004
4	Demand Forecasting for Managers	Stephan Kolassa and Enno Siemsen.	Business Expert press	1st Ed., 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Industrial Engineering	Avraham Shtub, Yuval Cohen	Taylor and Francis	2nd Ed., 2017
3	Production and operations management	R B Khanna	PHI	2nd Ed., 2015
4	Modern production/operations management	Buffa and Sarin	John Wiley and Sons	8th Ed., 2007

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107292/
	https://archive.nptel.ac.in/courses/110/107/110107141/
	https://archive.nptel.ac.in/courses/106/105/106105195/

SEMESTER: S6

ADDITIVE MANUFACTURING

Course Code	OEPET 613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the students on various aspects and processes of additive manufacturing technology.
2. To provide a deeper insight on the applications of additive manufacturing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Additive manufacturing (AM): Introduction, need for AM, evolution of AM technologies, steps in AM, types of AM processes, ASTM standard process classification, applications, economic/environmental/societal benefits and impact, industry trends, opportunities and challenges, CAD design for AM, support structures.	9
2	Vat photopolymerization processes: Stereolithography (SLA), digital light processing (DLP) and masked stereolithography (MSLA), material jetting and binder jetting, applications. Extrusion based AM Processes: Principles of melt extrusion and deposition, fused filament fabrication (FFF), fused deposition modelling (FDM) and direct ink writing (DIW), applications.	9
3	Direct energy deposition (DED) processes: Laser engineered net shaping (LENS) and direct metal deposition (DMD), applications. Powder bed fusion (PBF) processes: Direct metal laser sintering (DMLS), selective laser sintering (SLS), electron beam melting (EBM), selective laser melting (SLM) and multijet fusion (MJF), materials and applications.	9
4	Laminated object manufacturing (LOM): Ultrasonic consolidation (UC), applications.	9

	AM defects: Lack of fusion, keyhole collapse, gas porosity, solidification cracking, solid-state cracking, and surface-connected porosity. Materials science for AM: Multifunctional and functionally graded materials in AM. Case studies of AM applications: Medical, aerospace, automotive, and consumer products.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concepts, processes and applications of AM technology.	K2
CO2	Demonstrate vat polymerisation, extrusion and direct energy deposition techniques in AM.	K2
CO3	Compare the key aspects of powder bed based and laser based AM techniques.	K2
CO4	Explain LOM technique, material science of AM and various defects in AM; apply various processes of AM in industries.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	2	2	-	-	-	-	2
CO2	3	-	-	-	-	1	-	-	-	-	-	2
CO3	3	-	-	-	-	1	-	-	-	-	-	2
CO4	3	-	2	-	-	1	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Additive Manufacturing Technologies	Ian Gibson, David W. Rosen, Brent Stucker, Mahyar Khorasani	Springer	3rd Ed., 2021
2	Additive Manufacturing :Principles, technologies and Applications	C.P Paul ,A N Jinoop	McGrawHill	1st Ed., 2021
3	Additive Manufacturing Technologies	S Shiva, Anuj K Shukla	Wiley	1st Ed., 2024

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Additive Manufacturing and 3D Printing Technology: Principles and Applications	G K Awari, C S Thorat, Vishwajeet Ambade, D P Kothari	CRC Press	1st Ed., 2021
2	3D Printing and Additive Manufacturing : Principles and Applications	Chee Kai Chua, Kah Fai Leong	World Scientific	5th Ed., 2017
3	Additive Manufacturing: A Tool for Industrial Revolution 4.0	M. Manjaiah K, Raghavendra N, Balashanmugam, J Paulo Davim	Elsevier	1st Ed., 2021
4	Additive Manufacturing Technology: Design, Optimization and Modelling	Kun Zhou	Wiley -VCH	1st Ed., 2022
5	Additive Manufacturing Handbook: Product Development for the defence Industry	Adedeji B Badiru, Vhance V Valencia, David Liu	CREC press	1st Ed., 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/103/112103306/
	https://archive.nptel.ac.in/courses/112/104/112104312/

SEMESTER: S6

INDUSTRIAL AUTOMATION AND ROBOTICS LAB

Course Code	PCPEL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To provide students with hands-on experience in industrial automation.
2. To train students in robot programming for industrial production.

Expt. No.	Experiments (Minimum 10 Experiments should be completed)
1	Pneumatic circuit simulation with single acting double acting cylinders.
2	Sequencing of multiple cylinder actuation in different combinations - pneumatic.
3	Sequencing of multiple cylinder actuation in different combinations - electro-pneumatic.
4	Automatic inspection systems using machine vision.
5	Sensor based process monitoring and control.
6	PLC Programming for process automation.
7	Pick and place robot operation using a 6 axis arm.
8	Coordinated motion of multiple actuator, electro - pneumatic systems in a desired sequence using hard - wire programmed control systems.
9	Coordinated motion of multiple actuator, electro - pneumatic systems in a desired sequence using PLC.
10	Interfacing of an LVDT with a PC for monitoring the displacement of machine slides and raising an alarm if the displacement exceeds specified limit.
11	Control of speed, direction and number of revolutions of a stepper motor using a PC.
12	Determination of maximum and minimum position of links. Estimation of accuracy, repeatability and resolution.
13	Verification of transformation (Position and orientation) with respect to gripper and world coordinate system.
14	Robot programming and simulation for Colour identification.
15	Open and closed loop control of AC and DC drives.
16	Characterize the temperature sensor (RTD) and LVDT.
17	Simulation of Mathematical Model of Robot.
18	Forward and Inverse Dynamic Analysis of a 2-DOF Robotic Manipulator using Software Tools.

19	Building and Programming a Simple Arduino-Based Robot for basic movement.
20	Write a simple program to control a robot's motion (e.g., moving forward, backward, turning) using programming commands.
21	Development of an obstacle avoidance robot using servo motors, ultrasonic and touch sensors.
22	Design of conveyor automation system using PLC, SCADA and Electrical drive.

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Model and simulation of automated systems using pneumatics.	K3
CO2	Model and simulation of automated systems using electro-pneumatics.	K3
CO3	Develop Program for robots in industrial application.	K3
CO4	Develop automated industrial systems using sensors and image capture.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	3	2	-	-	2	2	-	3
CO2	3	-	2	-	3	2	-	-	2	2	-	3
CO3	3	-	2	-	2	2	-	-	2	2	-	3
CO4	3	-	3	-	2	2	-	-	2	2	-	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of pneumatic control engineering	J P Hasebrink and R Kobler	Festo Didactic	3rd Ed., 2000
2	Robot operating system vol 1&2	Anis Kouba	Springer	2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Sherlock Machine vision software	Teledyne Dalsa	Teledyne Dalsa Industrial products	2017
2	Open GL 3 programming guide	Dan Ginsburg), Budirijanto Purnomo , Dave Shreiner & Aaftab Munshi	Addison-Wesley	2nd Ed., 2014

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

PRODUCTION ENGINEERING

SEMESTER: S7

ADVANCED WELDING AND METAL JOINING TECHNIQUES

Course Code	PEPET741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the principles and applications of advanced welding and metal joining techniques.
2. To equip students with the latest developments and innovations in welding and metal joining.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of metal joining process: Importance of advanced joining techniques, applications in industries. Solid state welding: Friction welding, friction stir welding, inertia welding-working principle and applications. Explosive welding, diffusion bonding, high frequency induction welding-working principles, process parameters and applications. Plasma arc welding:Theory and principles, transferred and non - transferred arc techniques, equipment and tooling, electroslog welding, electroslog welding.	9
2	Radiant energy welding: Electron beam welding, background of the process and welding equipment, laser beam welding, physics of lasers, types of lasers, process parameters, applications and limitations, hybrid laser - arc welding. ultrasonic welding - principles of operations, process characteristics and applications.	9
3	Inert gas arc welding: Gas metal arc welding (GMAW), Gas tungsten arc welding (GTAW) - equipment used, working principles. heat affected zone and its characteristics, Metallurgy of welding: Weldability of steels, cast iron, stainless steel, Mg, Cu, zirconium and titanium alloys, underwater welding process - equipment	9

	used, applications. Adhesive joining: Types of adhesives, joining of dissimilar materials, applications.	
4	Magnetic pulse welding: High speed welding using magnetic pulse. microwave welding - high speed welding using microwave energy. Modern soldering and brazing technique: Induction, dip resistance, diffusion process - hot gas wave and vapour phase soldering. Automated welding using robots: Overview of automation of welding, welding applications in aerospace, nuclear, surface transport vehicles. Joining of advanced materials: Joining of composites, joining of smart materials, joining of nanomaterials.	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the principles and applications of various solid state welding processes, electrosag and electro gas welding.	K2
CO2	Demonstrate the principles and process of Radiant Energy welding.	K2
CO3	Apply the weldability of materials at different contexts of industrial applications, and explain the inert gas arc welding welding and adhesive joining of dissimilar materials.	K3
CO4	Demonstrate the magnetic pulse welding process and modern soldering; utilise automated welding processes for various applications.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	2	2	-	-	-	-	-	2
CO2	3	-	-	-	2	2	-	-	-	-	-	2
CO3	3	-	2	-	2	2	-	-	-	-	-	2
CO4	3	-	2	-	2	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Welding Technology	Howard B Cary, Scott C Helzer	Pearson/Scott C. Helzer	6th Ed., 2005
2	Welding Engineering and Technology	Dr. R S Parmar	Khanna Publishers	3rd Ed.. 2013
3.	Advanced Welding Process	J N DuPont and J M Sanders	CRC Press.	2nd Ed., 2019
4.	Welding Metallurgy and Weldability	John C Lippold	John Wiley & Sons	4th Ed., 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Welding Brazing & Soldering	ASM Handbook	ASM Handbook	6th Ed., 2010
2	A Textbook of Welding Technology	O P Khanna	Dhanpat Rai Publications	1st Ed., 2005
3	Welding Engineering.	N K Sreenivasan	Tata McGraw-Hill Education.	4th Ed., 2004
4	“Welding: Principles and Applications”	Larry Jeffus	Cengage Learning	8th Ed., 2019
5	“Automation and Robotics in Welding”	S J Mistry	Taylor & Francis	1st Ed., 2017
6	“Welding Automation in Aerospace Industry”	S Kumar	CRC Press	1st Ed., 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107213/
	https://archive.nptel.ac.in/courses/112/103/112103305/

SEMESTER: S7

SIX SIGMA

Course Code	PEPET742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the tools and techniques of six sigma quality philosophy.
2. To apply DMAIC methodology for quality improvement.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Quality fundamentals and key concepts: Quality concepts and definition, history of continuous improvement, six sigma principles and focus areas, DMAIC methodology, six sigma applications. Quality management: Basics and key concepts, fundamentals of total quality management, cost of quality, voice of customer, quality function deployment (QFD), management and planning tools.	9
2	Define phase: Six sigma project identification, selection and definition, project charter and monitoring, process characteristics and analysis, process mapping SIPOC. Measure phase: Data collection and summarization, measurement systems fundamentals, measurement systems analysis, Gage R&R study, fundamentals of statistics, probability theory, process capability analysis, measures and indices, non-normal process capability analysis.	9
3	Analyze phase: Hypothesis testing fundamentals, hypothesis testing single population test, hypothesis testing two population test, correlation and regression analysis, one-way ANOVA, two-way ANOVA, failure mode and effect analysis (FMEA). Improve phase: Introduction to design of experiment, randomized block	9

	design, randomized block design, factorial design, Taguchi method - key concepts and illustrative applications.	
4	Control phase: Seven QC tools, statistical process control, control charts for variables, control charts for attributes. Six sigma implementation challenges: Design for six sigma (DFSS) DMADV, DMADOV, DFX, team management, case studies.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of quality and quality management.	K2
CO2	Identify and summarise the define and measure phases of Six Sigma.	K3
CO3	Select appropriate tools to analyse and improve processes.	K3
CO4	Select the suitable tools to control processes and identify the implementation challenges of Six Sigma.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	2	-	-	-	-	-	-	-	-	2
CO2	3	-	3	-	2	2	-	-	-	-	-	2
CO3	3	-	3	-	2	2	-	-	-	-	-	2
CO4	3	-	3	-	2	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Six Sigma Handbook: A Complete Guide for Green Belts, Black Belts and Managers at All Levels	Thomas Pyzdek and Paul Keller	McGraw-Hill Education	5th Ed., 2023
2	The certified six sigma Green Belt Handbook	Roderick A. Munro and Govindarajan Ramu and Daniel J Zrymiak	ASQ Quality Press and Infotech Standards India Pvt Ltd	2nd Ed., 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Certified Six Sigma Black Belt Handbook	T. M. Kubiak and Donald W Benbow	Pearson Publication.	3rd Ed., 2018
2	Implementing Six Sigma	Forrest W Breyfogle III	John Wiley & Sons INC	2nd Ed., 2003
3	An Introduction to Six Sigma and Process Improvement	Evans J R and W M Lindsay	CENGAGE Learning	2nd Ed., 2015
4	Six Sigma for Green Belts and Champions	Howard S. Gitlow and David M Levine	Pearson Education Inc	1st Ed., 2004
5	Design and Analysis of Experiments	Montgomery D C	Wiley	10th Ed., 2019
6	Fundamentals of Quality Control and Improvement	Mitra Amitava	Wiley India Pvt Ltd	4th Ed., 2016
7	Statistical Quality Control A Modern Introduction	Montgomery D C	Wiley	7th Ed., 2012
8	Lean Six Sigma and Minitab: The Complete Toolbox Guide for All Lean Six Sigma Practitioners	Quentin Brook	OPEX Resources	7th Ed., 2022

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105123/
	https://archive.nptel.ac.in/courses/110/105/110105039/

SEMESTER: S7**DESIGN OPTIMIZATION**

Course Code	PEPET743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the standard methods of mathematical modeling techniques in design.
2. To enable students to apply advanced modeling techniques in practical scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to design: System design, morphology of design, concept of workable design, practical example on workable system and optimal design. Newton Raphson method, one unknown - examples, multiple unknowns - examples. Gauss Seidel method - examples, rudiments of finite difference method for partial differential equations with example.	9
2	Regression and curve fitting: Need for regression in simulation and optimization, concept of best fit and exact fit. Exact fit - Lagrange interpolation, Newton's divided difference - examples. Least square regression - theory, Gauss Newton method for nonlinear least squares regression - examples.	9
3	Optimization and algorithms: Method of steepest ascent/steepest descent, conjugate gradient method, power law forms, geometric programming. Optimization algorithms, complex R-F, genetic algorithms, multi objective optimization, priori methods, posteriori methods.	9
4	Advanced optimization: Interactive methods, interpreting the Pareto front, multidisciplinary optimization, multidisciplinary feasible MDF, bi-level integrated system synthesis (BLISS), surrogate models, types - polynomial response surfaces, kriging, neural networks, radial basis functions - case studies.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the system concept in design and apply numerical techniques in system design.	K3
CO2	Develop relationships between different design variables using correlation and curve fitting.	K3
CO3	Solve multi objective design problems using optimization methods and algorithms.	K3
CO4	Solve multidisciplinary design optimization problems for various industrial applications	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	-	-	-	2
CO2	3	3	2	-	-	-	-	-	-	-	-	2
CO3	3	3	2	-	-	-	-	-	-	-	-	2
CO4	3	3	3	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Essentials of Thermal System Design and Optimization	C Balaji	CRC Press	1st Ed., 2018
2	System design optimization for product manufacturing	Masataka Yoshimura	Springer-Verlag London Limited	1st Ed., 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to optimum design	J S Arora	McGraw Hill	4th Ed., 2017
2	Optimization for engineering design - algorithms and examples	K Deb	Prentice Hall	2nd Ed., 2012
3	Design and optimization of thermal systems	Y Jaluria	CRC Press	2nd Ed., 2007
4	Elements of thermal fluid system design	L C Burmeister	Pearson	1st Ed., 1998
5	Design of thermal systems	W F Stoecker	McGraw Hill	3rd Ed., 2011

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/111/107/111107105/
	https://archive.nptel.ac.in/courses/106/108/106108056/

SEMESTER: S7

ERGONOMICS

Course Code	PEPET744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the concepts of ergonomics, work induced stress and fatigue.
2. To enable the students to apply the concepts of work environment and workplace design in industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Ergonomics: Introduction, understanding of ergo-system, fitting task to man their contractual structure, humanise work, basic body mechanics, anatomy of spine and pelvis for postures, stability and postural adaptation, musculoskeletal disorders in work spaces, behavioural aspects of postures, sources of human variability, principles of applied anthropometry, application of anthropometry in design, design for everyone, personal space.	9
2	Work induced stress and fatigue: Design of standing and sitting, workstation design, work surface design, visual display units, guidelines for design of static and repetitive works, ergonomic interventions - handle design, keyboard design. Anatomy and biomechanics of manual handling, prevention of injuries in manual handling, design of manual handling task, carrying, muscle structure - function and capacity, factors affecting physical work capacity.	9
3	Work environment and their interactions: Eye Vision, measurement of light, lighting design consideration, design of visual displays, visual fatigue, eye strain and near work, indoor lighting, psychological aspects. Measurement of sound, ear protection, design of acoustic environment, industrial noise control, design of visual displays, design of audio displays, design of controls, combining displays and controls, virtual environment.	9

4	Applied physiology in work space: Measurement of physiological cost of work, indoor climate, protection against extreme climates. Human error and equipment design, mental workload in human machine interaction, characteristics of human machine interaction, prevention of human error, human computer interaction, human centred design process for interactive systems, human computer dialogue, system design methods for ergonomics, cross cultural considerations.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Interpret ergonomics and principles of anthropometry.	K2
CO2	Explain work induced stress and fatigue.	K2
CO3	Apply the work environment and their interactions in workplace design.	K3
CO4	Demonstrate the work physiology and its application in work places.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	-	2	2	-	-	-	-	2
CO2	3	-	2	-	-	2	2	-	-	-	-	2
CO3	3	-	2	-	-	2	2	-	-	-	-	2
CO4	3	-	2	-	-	2	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Ergonomics	Bridger R S	McGraw Hill	1st Ed., 1995
2	Human Factors in Engineering and Design	McCormick J	McGraw Hill	1st Ed., 1992

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Factors in Product Design	Green W S and Jordan P W	Taylor & Francis	1st Ed., 2019
2	A Guide to Human Factors and Ergonomics	Martin Helander	CRC Press	2nd Ed.2005
3	Ergonomic Workplace Design for Health, Wellness, and Productivity	Alan Hedge	CRC Press	1st Ed., 2016

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/104/112104222/

SEMESTER: S7

ENERGY MANAGEMENT AND AUDIT

Course Code	PEPET745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the principles and concepts of energy management and auditing.
2. To equip students to practice energy management and auditing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to energy management: Overview of energy management and auditing, energy scenario in various sectors, energy consumption in various areas. Energy policies and regulations: BEE schemes under energy conservation act, electricity act 2003-I, electricity act 2003-II, integrated energy policy-I, integrated energy policy-II.	9
2	Energy management and energy audit: Objectives, energy management frameworks and standards, energy audit and its types, energy audit procedures and checklist, energy performance and benchmarking.	9
3	Implementation of energy audit: Instruments involved in energy auditing, introduction to monitoring and targeting, analysis of information and data with management information system, planning of energy policy, evaluation and implementation of energy performance, tools for effective implementation.	9
4	Case studies in energy management and audit: Real life case studies of energy auditing and management, developing an energy management plan, energy saving measures and projects, environment impact assessment.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> . <i>Clearly defines the real-world quality issue.</i> . <i>Examine and identify relevant contextual factors.</i>	5
	<i>Problem Analysis</i> . <i>Present a structured realistic solution methodology.</i> . <i>Compare and justify the proposed solutions with evidence and logical reasoning</i>	5
Evaluate	<i>Validation of Results</i> . <i>Thoroughly evaluate the proposed solutions.</i> . <i>Compares trade-offs, advantages, and disadvantages.</i> . <i>Considers feasibility, scalability, and practical implications.</i>	5
	<i>Conclusion and Report Writing</i> . <i>Summarizes procedure, findings and insights, limitation, and scope for future work.</i> . <i>Preparation of Report with all components of project report.</i>	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basics of energy management and its policies.	K2
CO2	Demonstrate energy audit and energy performance benchmarking.	K2
CO3	Develop skills in analysing energy data and making recommendations.	K3
CO4	Analyse energy audit in real life situations.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	2	-	-	-	-	-	-	2
CO2	3	-	-	-	2	-	-	-	-	-	-	2
CO3	3	-	2	-	2	-	-	-	-	-	-	2
CO4	3	-	3	-	2	-	3	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Energy Conservation and Audit	Agarkar Santosh Vyankatro and Mateti Naresh Kumar	Lambert Academic publishing	1st Ed., 2015
2	Energy Management, Audit and Conservation	Barun Kumar De	Vrinda publication	1st Ed., 2007
3	Energy Management and Conservation	K V Sharma and P Venkata Seshiah	I K International Publishing House	1st Ed., 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Energy Audits	Albert Thumann and Terry Niehus.	CRC press	9th Ed., 2013
2	Guide to Energy Management	Barney L. Capehart, Wayne C. Turner, and William J. Kennedy.	CRC press	8th Ed., 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://onlinecourses.swayam2.ac.in/nou23_es05
	https://archive.nptel.ac.in/courses/124/107/124107160/

SEMESTER: S7

ADVANCED MATERIALS AND PROCESSES

Course Code	PEPET751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the characteristics of structural and functional materials
2. To utilize the advanced techniques used in selecting materials for manufacturing

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Structural and functional materials: Introduction and classification of energy materials and smart materials, evolution of engineering materials, metastable alloys. High temperature Materials: Structure, processing, mechanical behaviour and oxidation resistance of stainless steels, Ni- and Co- based superalloys, aluminides and silicides, carbon-carbon and ceramic composites, biomaterials.	9
2	Shape-Memory Alloys: Mechanisms of one-way and two-way shape memory effect, reverse transformation, thermoelasticity and pseudo-elasticity, examples and applications. Bulk Metallic Glass: Criteria for glass formation and stability, examples and mechanical behaviour.	9
3	Nano-materials: Classification, size effect on structural and functional properties, processing and properties of nano-crystalline materials, thin films and multilayered coatings. Single walled and multiwalled carbon nanotubes: Methods of synthesis of nanomaterials, applications.	9
4	Soft and hard magnetic materials for storage devices: Design and processing, piezoelectric materials - processing and properties.	9

	Advanced processes applied for advanced materials: Single crystal growth, rapid solidification, inert gas condensation, physical and chemical vapour deposition of thin films.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain structural and functional materials for various industrial applications.	K2
CO2	Demonstrate the characteristics and applications of shape-memory alloys and bulk metallic glass.	K2
CO3	Explain nano-materials and applications	K2
CO4	Illustrate advanced processes applied for advanced materials	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	2	2	-	-	-	-	2
CO2	3	2	2	-	-	2	2	-	-	-	-	2
CO3	3	2	2	-	-	2	2	-	-	-	-	2
CO4	3	2	2	-	-	2	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Engineering and Technology	Serope Kalpakjian and Steven R Schmid	Addison Wesley Longman (Singapore) Pvt. Ltd., New Delhi	5th Ed., 2000
2	Nanoengineering of structural, functional and smart materials	Mark J. Schulz	Taylor & Francis	1st Ed., 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nanocomposite science and technology	Pulikel M. Ajayan	Wiley-VCH	1st Ed., 2005
2	Analysis and Performance of Fiber Composites	B. D. Agarwal, L. J. Broutman	John Wiley	4th Ed., 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/noc/courses/noc18/SEM2/noc18-mm12/

SEMESTER: S7

ADDITIVE MANUFACTURING TECHNOLOGIES

Course Code	PEPET752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise various aspects of additive manufacturing such as choice of materials, technology and process capability of each process.
2. To enable students to apply the design and applications in additive manufacturing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Need for additive manufacturing (AM), evolution of AM technologies, steps in AM, types, AM processes, ASTM standard process classification, applications of AM in medical, aerospace, automotive, and consumer products, benefits and societal impacts, industry trends, opportunities and challenges. AM design process: CAD design for AM, STL File generation, staircase effect, support structure design and removal, anisotropy of print parts.	9
2	Vat photopolymerisation techniques: Stereolithography (SLA), digital light processing (DLP) and masked stereolithography (MSLA). Resin materials - properties, types and applications, material jetting and binder jetting, process capabilities and applications. Extrusion based AM processes: Principles of melt extrusion and deposition, techniques - fused filament fabrication (FFF), fused deposition modelling (FDM) and direct ink writing (DIW), material properties, process capability, applications.	9
3	Powder bed fusion based AM processes: Direct metal laser sintering (DMLS), selective laser sintering (SLS), electron beam melting (EBM), selective laser melting (SLM) and multijet fusion (MJF), process	9

	phenomena, powder handling, process capabilities, comparison, materials. SLM process: Laser material interaction, effect of laser parameters, melting modes - conduction and keyhole mode of operation, marangoni effect. Laminated object manufacturing (LOM): Ultrasonic consolidation (UC) process capability, applications.	
4	Direct energy deposition (DED) based AM Processes: Laser engineered net shaping (LENS), direct metal deposition (DMD), process capabilities and applications. AM defects: Prediction and mitigation, post-processing techniques. Materials science for AM: Multifunctional and functionally graded materials in AM process, structure, property correlation. Design for additive manufacturing (DfAM): DfAM guidelines, design of lightweight and porous structures.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain AM technology and its processes, design and printing of AM components.	K2
CO2	Demonstrate vat and extrusion based deposition techniques in AM.	K2
CO3	Interpret the key differences of powder bed and DED based AM processes; explain SLM process and LOM.	K2
CO4	Apply DfAM concepts in lightweight designs.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	1	2	-	-	-	-	2
CO2	3	-	-	-	-	1	-	-	-	-	-	2
CO3	3	-	2	-	-	1	-	-	-	-	-	2
CO4	3	-	2	-	-	1	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Additive Manufacturing Technologies	Ian Gibson, David W. Rosen, Brent Stucker, Mahyar Khorasani	Springer	3rd Ed., 2021
2	Additive Manufacturing :Principles, technologies and Applications	C.P Paul ,A N Jinoop	McGrawHill	1st Ed., 2021
3	Additive Manufacturing Technologies	S Shiva ,Anuj K Shukla	Wiley	1st Ed., 2024

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Additive Manufacturing and 3D Printing Technology: Principles and Applications	G K Awari, C S Thorat, Vishwajeet Ambade, D P Kothari	CRC Press	1st Ed., 2021
2	3D Printing and Additive Manufacturing : Principles and Applications	Chee Kai Chua, Kah Fai Leong	World Scientific	5th Ed., 2017
3	Rapid Prototyping: Laser-based and Other Technologies	Patri K Venuvinod , Weiyin Ma	Springer Verlag	1st Ed., 1998
4	Additive Manufacturing: A Tool for Industrial Revolution 4.0	M. Manjaiah K, Raghavendra N, Balashanmugam, J Paulo Davim	Elsevier	1st Ed., 2021
5	Rapid Prototyping: Principles and Applications in Manufacturing	Rafiq Noorani	Wiley	1st Ed., 2006
6	Additive Manufacturing Technology: Design, Optimization and Modelling	Kun Zhou	Wiley -VCH	1st Ed., 2022
7	Additive Manufacturing Handbook: Product Development for the defence Industry	Adedeji B Badiru, Vhance V Valencia, David Liu	CREC press	1st Ed., 2017
8	Additive Manufacturing of Metals : From Fundamental technology to Rocket Nozzles,Medical Implants and custom jewellery	John O Milewski	Springer	1st Ed., 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/103/112103306/
	https://archive.nptel.ac.in/courses/112/104/112104312/

SEMESTER: S7

PRODUCT AND BRAND MANAGEMENT

Course Code	PEPET753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the concepts and principles of product and brand management.
2. To equip students to apply the concepts of product life cycle, product demand estimation, product development and launching.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Product management: Introduction, terminologies associated with product, product planning, product classification, product differentiation and its elements. product category analysis, insight, innovation and design thinking. New product development: Factors affecting the success and failure of new product development. Product life cycle: concept, strategies related to different stages of PLC.	9
2	Market and product planning: Product strategy and decisions, product pricing, product packaging, meaning, importance, types, features, factors influencing packing decision, green packing - concept and importance, product distribution and promotion strategies.	9
3	Brand Management: Meaning, functions and significance-types of brands, concept of branding. Branding strategies, branding challenges and opportunities, brand equity concept, customer based brand equity, brand switching. Measuring and interpreting brand performance: Developing a brand equity measurement and management system, the brand value chain, designing brand tracking studies.	9
4	Brand architecture strategies: Brand portfolio, brand hierarchy, brand extensions, co-branding, corporate branding, brand management tools, brand communication-need and advantages of brand communication. Brand loyalty and equity-factors affecting brand loyalty, brand equity, brand extension.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Develop effective product strategies, plans and Analyse market trends and customer needs across the PLC	K3
CO2	Apply the concepts of packing materials, sustainability and environmental impacts.	K3
CO3	Apply the brand management concepts for developing brand strategies.	K3
CO4	Develop brand architecture strategies for different marketing environments.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	-	-	-	-	-	-	-	2
CO2	3	-	2	-	-	-	2	-	-	-	-	2
CO3	3	-	2	-	-	-	-	-	-	-	-	2
CO4	3	-	2	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The new strategic Brand Management: Creating and sustaining Brand Equity.	J.N Kapferer	Kogan Page	4th Ed., 2004
2	Strategic Brand Management	Kevin Lane Keller	Pearson Education	5th Ed., 2020
3.	Product and brand Management	David A. Aaker, Robert Jacobson.	Wiley	7th Ed., 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product and brand management	Tapan panda	Oxford University Press	1st Ed., 2016
2	Brand Management	Harsh V Verma	Excel	3rd Ed., 2013
3	Brand Management Principles and Practice	Kirti Dutta	Oxford University Press	2nd Ed., 2015
4	Product Management	Don Seller, Meri Gruber	Routledge	2nd Ed., 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/107/110107161/ https://archive.nptel.ac.in/courses/112/107/112107238/

SEMESTER: S7

HEAT AND MASS TRANSFER

Course Code	PEPET754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart the concept of various modes of heat and mass transfer.
2. To enable students to apply the modes of heat and mass transfer rates in conduction, convection and radiation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Modes of heat transfer: Conduction, convection and radiation, Fourier law, Newton's law, Stefan Boltzmann law, thermal resistance, analogy between flow of heat and electricity, general differential equation, cartesian coordinate, thermal diffusivity. One dimensional steady state heat conduction: Plane and composite systems (plane wall, cylinder and sphere), importance of critical thickness, conduction with internal heat generation, extended surfaces.	9
2	Convection: Free and forced convection, boundary layer concepts, forced convection, external flow-flow over plates, cylinders, sphere, bank of tubes, internal flow, free convection-flow over vertical plate, horizontal plate, inclined plate, cylinder and sphere.	9
3	Radiation: Radiation laws and radiate properties, black body and gray body radiation, radiosity, view factor relations, electrical analogy. radiation shields. Boiling and condensation: Film wise and drop wise condensation.	9
4	Mass transfer: Diffusion mass transfer, Fick's law of diffusion, equimolar diffusion, diffusion coefficient, convective mass transfer, analogy with heat transfer, convective mass transfer correlations.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic principles of heat transfer and apply convection mode of heat transfer.	K3
CO2	Make use of the concepts in convection mode of heat transfer for solving industrial problems.	K3
CO3	Apply radiation mode of heat transfer to solve industrial problems.	K3
CO4	Make use of the concepts in mass transfer to solve practical problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	2	-	-	-	-	-	2
CO2	3	3	-	-	-	2	-	-	-	-	-	2
CO3	3	3	-	-	-	2	-	-	-	-	-	2
CO4	3	3	-	-	-	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A text book of heat and mass transfer	R. K Rajput	S Chand	Revised edition 2018
2	Fundamentals of heat and mass transfer	R C Sachdeva	New age science	4th Ed., 2009
3	Heat and mass transfer	Yunus A Cenegal	McGraw Hill	5th Ed., 2017
4	Heat and mass transfer data book	C P Kothadaraman	New age international	10th Ed., 2022

Reference Books				
Sl.No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Heat Transfer	J P Holman	McGraw Hill	10th Ed., 2017
2	Heat and mass transfer	D S Kumar	S K Kataria & Sons	Reprint 2013
3	Principles of mass transfer	K Sharma	Prentice hall India	1st Ed., 2007

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/103/105/103105140/
	https://archive.nptel.ac.in/courses/112/101/112101097/

SEMESTER: S7

DESIGN FOR MANUFACTURING AND ASSEMBLY

Course Code	PEPET755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize with various product manufacturing design techniques.
2. To equip students with the basics of assembly techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to DFMA: History of DFMA, steps for applying DFMA during product design, advantages of applying DFMA during product design, reasons for not implementing DFMA. Materials and manufacturing: Introduction to materials and material selection, classification of engineering materials, material selection for product design.	9
2	Design for casting: Sand casting - typical characteristics of a sand cast part, design recommendation for sand casting. Investment casting - design consideration of investment casting, typical characteristics and applications. Die casting - applications, suitable material consideration for die casting, general design consideration, specific design recommendation. Injection moulding - introduction to injection moulding, typical characteristics of injection moulded parts, effect of shrinkage, suitable materials, design recommendations.	9
3	Design for machining: Introduction to machining, recommended materials for machinability, design recommendations, design for turning operation - process description, typical characteristics and applications, suitable materials, design recommendations. Machining round holes - introduction,	9

	suitable materials, design recommendations, recommended tolerances, parts produced by milling- process description, characteristics and applications of parts produced on milling machines, design recommendations for milling.	
4	Design for joining processes : Welding - design for recommendation for welding process. Solder and braze assembly - process, typical characteristics, suitable materials, detailed design recommendations. Introduction to Assembly: The assembly process, characteristics and applications, example of common assembly, economic significance of assembly, general taxonomies of assembly operation and systems, assembling a product. Design for assembly: Introduction, Design consideration, design for fasteners - introduction, design recommendation for fasteners.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> . <i>Clearly defines the real-world quality issue.</i> . <i>Examine and identify relevant contextual factors.</i>	5
	<i>Problem Analysis</i> . <i>Present a structured realistic solution methodology.</i> . <i>Compare and justify the proposed solutions with evidence and logical reasoning</i>	5

Evaluate	<i>Validation of Results</i> <i>Thoroughly evaluate the proposed solutions.</i> <i>Compares trade-offs, advantages, and disadvantages.</i> <i>Considers feasibility, scalability, and practical implications.</i>	5
	<i>Conclusion and Report Writing</i> <i>Summarizes procedure, findings and insights, limitation, and scope for future work.</i> <i>Preparation of Report with all components of project report.</i>	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the systematic design procedures with due consideration to the process and material constraints.	K2
CO2	Develop geometrical shapes of components with due consideration to the constraints of various metal casting processes.	K3
CO3	Select dimensional aspects of components with due consideration to the inherent process capabilities of various metal cutting processes.	K3
CO4	Analyse product architecture and material combination for components enabling an easier assembly process.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	-	-	-	-	-	2
CO2	3	-	3	-	-	-	-	-	-	-	-	2
CO3	3	-	3	-	-	-	-	-	-	-	-	2
CO4	3	-	3	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Design, Materials and Manufacture Guide	J Lesko	John Willy and Sons, Inc	2nd Ed., 2008
2	Engineering Design	George E Dieter and Linda C Schmidt Fourth edition	McGraw-Hill companies, New York, USA	4th Ed., 2017
3	Product Design for Manufacture and Assembly	Geoffrey Boothroyd, Peter Dewhurst and Winston Knight	CRC press, Taylor & Francis, Florida, USA	3rd Ed., 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design for Manufacturing and assembly,	O Molloy, S Tilley and E A Warman	Chapman & Hall, London, UK.	1st Ed., 1998
2	Mechanical Assemblies: Their Design, Manufacture, and Role in Product Development	D E Whitney	Oxford University Press, New York	1st Ed., 2004

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/107/103/107103012/
	https://archive.nptel.ac.in/courses/112/101/112101005/

SEMESTER: S7

SIX SIGMA

Course Code	OEPET721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the tools and techniques of six sigma quality philosophy.
2. To apply the DMAIC methodology for practical applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Quality fundamentals and quality management: Quality concepts and definition, history of continuous improvement, six sigma principles and focus areas, six sigma applications, design for six sigma (DFSS). Quality management basics and key concepts, fundamentals of total quality management, cost of quality.	9
2	Define: Six sigma project identification, selection and definition, project charter and monitoring, process characteristics and analysis, process mapping SIPOC. Measure: Data collection and summarization, measurement systems fundamentals, measurement systems analysis gauge R&R study, fundamentals of statistics, probability theory, process capability analysis key concepts, process capability analysis measures and indices.	9
3	Analyze: Hypothesis testing fundamentals, hypothesis testing- single population test, two population test, correlation and regression analysis, one-way ANOVA. Improve: Introduction to design of experiment, randomized block design, factorial design, voice of customer, quality function deployment (QFD).	9
4	Control: Seven QC tools, statistical process control key concepts, control charts for variables and attributes, acceptance sampling key concepts,	9

	design of acceptance, sampling plans for attributes. Six sigma methodologies: DMADV, DMADOV, DFX, roles and responsibilities in six sigma, applications.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of quality and quality management.	K2
CO2	Identify and summarise the define and measure phases of Six Sigma	K3
CO3	Select appropriate tools to analyse and improve processes	K3
CO4	Select the suitable tools to control processes and apply six sigma methodologies.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	2	-	-	-	-	-	-	-	-	2
CO2	3	-	3	-	2	2	-	-	-	-	-	2
CO3	3	-	3	-	2	2	-	-	-	-	-	2
CO4	3	-	3	-	2	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Six Sigma Handbook: A Complete Guide for Green Belts, Black Belts and Managers at All Levels	Thomas Pyzdek and Paul Keller	McGraw-Hill Education	5th Ed., 2023
2	The certified six sigma Green Belt Handbook	Roderick A Munro and Govindarajan Ramu and Daniel J Zrymiak	ASQ Quality Press and Infotech Standards India Pvt Ltd	2nd Ed., 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Certified Six Sigma Black Belt Handbook	T M Kubiak and Donald W Benbow	Pearson Publication	3rd Ed., 2018
2	Implementing Six Sigma	Forrest W Breyfogle III	John Wiley & Sons INC	2nd Ed., 2003
3	An Introduction to Six Sigma and Process Improvement	Evans J R and W M Lindsay	CENGAGE Learning	2nd Ed., 2015
4	Six Sigma for Green Belts and Champions	Howard S. Gitlow and David M Levine	Financial Times/ Prentice Hall	1st Ed., 2004
5	Design and Analysis of Experiments	Montgomery D C	Wiley	10th Ed., 2019
6	Fundamentals of Quality Control and Improvement	Mitra Amitava	Wiley India Pvt Ltd.	4th Ed., 2016
7	Statistical Quality Control A Modern Introduction	Montgomery D C	Wiley	7th Ed., 2012

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105123/
	https://archive.nptel.ac.in/courses/110/105/110105039/

SEMESTER: S7

ENERGY MANAGEMENT AND AUDIT

Course Code	OEPET722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the fundamentals of energy management and auditing.
2. To enable students to practice energy management and auditing.

SYLLABUS

Module No.	Syllabus description	Contact Hours
1	Introduction to various types of energy: Energy scenario in various sectors, energy consumption in various areas, energy security and its conservation, energy and its various forms, basics of electricity, basics of thermal energy.	9
2	BEE schemes under energy conservation act: Electricity act 2003-I, electricity act 2003-II, integrated energy policy-I, integrated energy policy-II, objectives of energy management and energy audit, energy audit and its types, energy performance and benchmarking.	9
3	Implementation of energy audit: Instruments involved in energy auditing, introduction to monitoring and targeting, analysis of information and data with management information system, planning of energy policy, evaluation and implementation of energy performance, tools for effective implementation.	9
4	Energy and environmental issues: Impact of global warming in climate change, international agreements and panels on climate change, protocols and unions to combat climate change, sustainable development, introduction to renewable energy sources, fundamentals of solar, wind hydro power, biomass, wave tidal, geothermal energy, fuel cell and energy from wastes.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain various types of energy and its conservation.	K2
CO2	Demonstrate energy policies and performance benchmarking.	K2
CO3	Develop skills in analysing energy data and making recommendations	K3
CO4	Illustrate environmental impacts and sustainability of energy resources	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	2	-	-	-	-	-	-	2
CO2	3	-	-	-	2	-	-	-	-	-	-	2
CO3	3	-	2	-	2	-	-	-	-	-	-	2
CO4	3	-	-	-	2	-	3	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Energy Conservation and Audit	Agarkar Santosh Vyankatro and Mateti Naresh Kumar	Lambert Academic publishing	1st Ed., 2015
2	Energy Management, Audit and Conservation	Barun Kumar De	Vrinda publication	1st Ed., 2007
3	Energy Management and Conservation	K V Sharma and P Venkata Sessaiah	I K International Publishing House	1st Ed., 2007
4	Energy Engineering and Management	Chakrabarti A	PHI learning	2nd Ed., 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Energy Audits	Albert Thumann and Terry Niehus.	CRC press	9th Ed., 2013
2	Guide to Energy Management	Barney L. Capehart, Wayne C. Turner, and William J. Kennedy.	CRC press	8th Ed., 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://onlinecourses.swayam2.ac.in/nou23_es05
	https://archive.nptel.ac.in/courses/124/107/124107160/

SEMESTER: S7

BUSINESS TO BUSINESS MARKETING

Course Code	OEPET 723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise students with concepts of the B2B market and marketing strategies.
2. To gain knowledge about product management, pricing strategies and B2B marketing research.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Business to Business (B2B) marketing: Business marketing, classifying goods for the business market, market structure, environment and characteristics of business marketing, strategic role of marketing, B2B vs B2C (Business to consumer) marketing. Organizational buying and buyer behaviour: Organizational buyers' decision process, stepwise model and a process flow model, organizational and business markets, government as a customer, commercial and institutional customers, buygrid framework.	9
2	B2B Marketing Strategy: Strategy making and strategy management process, industrial product strategy, managing products for business markets-managing services for business markets, segmentation, targeting and positioning (STP). Business marketing communications: B2B advertising, digital marketing, trade shows, exhibitions and business meets. Industrial market forecasting: Demand forecasting models, qualitative and quantitative forecasting methods.	9
3	Product management: Role of product in the industrial market, new product development, industrial product life cycle, technology innovation life cycle,	9

	product evaluation matrix, techniques for identifying new products, quality function deployment (QFD), perceptual mapping, reverse engineering, fish bone diagram, role of service and maintenance in industrial markets, strategies for managing services in business markets.	
4	Pricing: Pricing strategies, the pricing policy, price on the internet, financial marketing, competitive bidding, commercial terms and conditions, buyer seller relationship, types of relationships, transactional and collaborative relationships, influencing industrial customers. B2B marketing research: Challenges in B2B research, developing a marketing information system, role of qualitative research techniques in B2B research, business marketing channels, implementing and controlling marketing plan.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain B2B marketing and Organizational buying and buyer behaviour.	K2
CO2	Demonstrate B2B marketing strategies that align with business objectives and business market needs.	K2
CO3	Make use of the role of product management within B2B organizations and its relationship with other departments.	K3
CO4	Demonstrate the pricing strategies and B2B marketing research.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	1	-	-	-	-	-	-	-	-	2
CO2	3	-	2	-	-	-	-	-	-	-	-	2
CO3	3	-	2	-	-	-	-	-	-	-	-	2
CO4	3	-	1	-	-	-	-	-	-	-	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Business Market Management Understanding, Creating and Delivering Value	James C. Anderson, Das Narayandas, James A. Narus and D.V.R. Seshadri	Pearson	3rd Ed., 2010
2	Business Marketing: Text and Cases	Krishna K Havaladar	Mcgraw Hill	3rd Ed., 2014
3	Business to Business Marketing Management: A Global Perspective	Alan Zimmerman, Jim Blythe	Routledge	4th Ed., 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Innovative B2B Marketing New Models, Processes and Theory	Simon Hall	Kogan Page	2nd Ed., 2022
2	Business-to-Business Marketing	Ross Brennan, Louise E. Canning	SAGE Publications	2nd Ed., 2011

Video Links (NPTEL, SWAYAM...)	
Link ID	https://nptel.ac.in/courses/110107168

SEMESTER 8

PRODUCTION ENGINEERING

SEMESTER: S8

DESIGN AND IMPLEMENTATION OF HUMAN COMPUTER INTERFACES

Course Code	PEPET861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the principles and concepts of human-computer interfaces and its implementation.
2. To equip the students to apply the concepts of human-computer interfaces and its design in practical scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to human-computer interfaces: Interactive systems, introduction to usability, engineering for usability, interactive system life cycle, usability requirements, contextual inquiry, functional requirements, specification, software requirements specifications (SRS), usability and non-functional requirements to SRS.	9
2	Introduction to interface design: Shneiderman's golden rules, Norman's principles, prototyping, prototype evaluation, case studies in prototype evaluation. Basics of system design: Data flow diagram (DFD), entity relationship diagram (ERD), case studies.	9
3	Object oriented design and coding: Unified modelling language (UML), coding basics, code testing basics, review-based code testing, case studies. Execution based code testing: Test cases and test suits, black - box testing - equivalence class partitioning and boundary value analysis approaches, white-box testing - control flow graph, case studies.	9

4	System Integration and usability analysis: System integration and testing - bottom-up testing and top down testing approaches, stubs and drivers, stages of testing- alpha testing, beta testing and acceptance testing, types of testing, empirical usability evaluation, experiment design, empirical data analysis, project management.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the principles, theories, and models of human-computer interaction.	K2
CO2	Make use of key principles of interface design, system design concepts in practical applications.	K3
CO3	Apply the coding concepts and its implementation in practical cases.	K3
CO4	Utilise the usability testing to evaluate interfaces, identifying usability issues and user preferences.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	3	2	-	-	-	-	-	2
CO2	3	-	3	-	3	2	-	-	-	-	-	2
CO3	3	-	3	-	3	2	-	-	-	-	-	2
CO4	3	-	3	-	3	2	-	-	-	-	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human-Computer Interaction: User-Centric Computing for Design	Samit Bhattacharya	McGraw Hill Education	1st Ed., 2019
2	Software Engineering: A Practitioner's Approach	Bruce R Maxim & Roger S Pressman	McGraw Hill Education	8th Ed., 2019
3	Designing the user interface: Strategies for effective human computer interaction	Ben Shneiderman, C Plaisant, M Cohen, S Jacobs	Pearson	5th Ed., 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Interaction Design: Beyond Human – Computer Interaction	Jennifer Preece, Helen Sharp, Yvonne Rogers	John Wiley & Sons	4th Ed., 2015
2	Human-Computer Interface Design	A G Sutcliffe	Springer	2nd Ed., 1995
3	Essential guide to User Interface design: An Introduction to GUI Design Principle and Techniques	W O Galitz	John Wiley & Sons	2nd Ed., 2002

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/103/106103237/

SEMESTER: S8

CORROSION / ENVIRONMENTAL DEGRADATION / SURFACE ENGINEERING

Course Code	PEPET862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise various modes of wear mechanisms, corrosion and the various surface engineering techniques.
2. To impart knowledge on surface degradation and failure analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Surface degradation mechanisms: Introduction, commercial perspective, adhesive wear mechanism, abrasive wear mechanism, wear phenomena defined by dimensionless parameters, surface deterioration, fracture surface degradation, heat and radiation on surface degradation, role of fatigue and fracture mechanisms in wear debris formation, friction and lubrication.	9
2	Erosive and cavitation wear: Fatigue, fretting, melting and diffusive wear mechanisms, brittle fracture wear, wear of ceramics and polymers. Corrosion: Electrochemical mechanism of corrosion, forms of corrosion, corrosion prevention and control, material selection for corrosion environments, corrosion inhibitors, corrosion testing and measurements, characterisation techniques.	9
3	Non - destructive testing: Recycling and reuse of materials, digitization of established curves and data driven approach. Life cycle assessment: Principal component analysis (PCA), failure / fault analysis, failure mode and effect analysis (FMEA), case studies.	9
4	Degradation mechanism and surface engineering: Synergistically combined	9

	degradation mechanism, fractography, environmental assisted surface deterioration, surface engineering, mitigation methods, types of maintenance and monitoring techniques, and terotechnology. Surface treatments: Surface coating, plating and anodizing, fusion processes, vapour phase processes, economic losses by corrosion degradation and its quantification.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the various modes of surface degradation mechanisms.	K2
CO2	Demonstrate the wear and corrosion mechanisms and its prevention.	K2
CO3	Explain the non-destructive testing techniques and life cycle analysis.	K2
CO4	Outline the surface degradation and mitigation methods.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	-	2	2	-	-	-	-	2
CO2	3	-	2	-	-	2	2	-	-	-	-	2
CO3	3	-	2	-	-	2	2	-	-	-	-	2
CO4	3	-	2	-	-	2	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Tribology, Friction and Wear of Engineering Material	I M Hutchings	Edward Arnold	1st Ed., 1992
2	Introduction to Tribology	B C Majumdar	S Chand	1st Ed., 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Tribology Handbook	Michael J Neale	Butterworth-Heinemann	2nd Ed., 1995
2	Maintenance Engineering Handbook	Darrin J Wikoff, Keith Mobley	McGraw-Hill	8th Ed., 2014
3	Corrosion & Corrosion Control	Uhlig H H	John Wiley Publishers	3rd Ed., 1985

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/113/105/113105105/

SEMESTER: S8

CONDITION MONITORING AND MAINTENANCE MANAGEMENT

Course Code	PEPET863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize maintenance concepts predictive maintenance in industries.
2. To enable students to plan and implement condition based maintenance methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Maintenance Concepts: Functions and objectives of maintenance, maintenance strategies, maintenance scheduling, organization and spare parts management, various methods and policies of maintenance engineering.	9
2	Predictive maintenance program: Condition based maintenance, condition checking and inspection and trend monitoring methods, routine inspection for pumps and compressors, preventive maintenance for rotating and stationary equipment, planning and scheduling of preventive maintenance.	9
3	Vibration analysis: Vibration parameters selection, classification of equipment, vibration severity charts, fault detection from vibration data, troubleshooting of mechanical equipment, wear debris analysis, oil analysis.	9
4	Reliability centered maintenance: Maintenance division models, reliability oriented maintenance systems, total productive maintenance (TPM) and benchmarking. procedures of JIT maintenance, zero defect maintenance and zero breakdown maintenance systems.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the procedures of maintenance management.	K2
CO2	Plan for predictive maintenance program in industries.	K3
CO3	Make use of condition monitoring techniques for fault detection.	K3
CO4	Develop maintenance procedures in line with a total productive maintenance approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	-	-	-	-	-	2
CO2	3	-	2	-	-	2	-	-	-	-	-	2
CO3	3	-	2	-	-	2	-	-	-	-	-	2
CO4	3	-	2	-	-	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Plant Maintenance and Reliability Engineering	Prof. N V S Raju	Cengage Learning India Publications	1st Ed., 2011
2	Engineering maintenance: A modern approach	B S Dhillon	CRC Press	1st Ed., 2002
3	Reliability Engineering: Probabilistic Models and Maintenance Methods	Joel A Nachlas	CRC Press	2nd Ed., 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Centered Maintenance – Reengineered	Jesus R. Sifonte, James V Reyes-Picknell	Productivity Press	1st Ed., 2017
2	Condition Monitoring in Large Thermal Power Plants : Power Plant Condition Monitoring	Shrikant Bhawe	Notions Press	2nd Ed., 2016

Video Links (NPTEL, SWAYAM...)	
Link ID	https://onlinecourses.swayam2.ac.in/nou21_me10/
	https://archive.nptel.ac.in/courses/112/105/112105232/

SEMESTER: S8

DYNAMICS OF MACHINERY

Course Code	PEPET864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the concepts of static and dynamic force analysis techniques and the principles of gyroscopic couple and balancing in dynamics.
2. To equip students to apply the concepts of single and multi-degree freedom vibrational behaviour to solve engineering problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Force analysis: Static force analysis-Freebody diagram - conditions for equilibrium - two and three force members, four force members - numerical examples involving single four bar chain and single slider crank chain. Dynamic force analysis: Inertia force and inertia torque D'Alembert's principle, force acting on the reciprocating parts of an engine, piston effort, force acting along the connecting rod, thrust on the sides of cylinder walls, crank pin effort and thrust on the crankshaft bearings - numerical examples.	9
2	Flywheel analysis: Flywheel, turning moment diagram, flywheel in different applications like IC engines, punching press - numerical examples. Gyroscopics: Gyroscopic couple and precessional motion, effect of gyroscopic couple on aeroplanes, ships - numerical examples. Balancing of rotating masses: Static and dynamic balancing, reference plane method for balancing of masses for different panes - analytical and graphical approach.	9
3	Longitudinal vibration: Single and two degree freedom system, overview of free vibration responses, equation of motion, underdamped, overdamped and critically damped response. Forced vibration: Frequency response curve, vibration isolation,	9

	transmissibility-numerical examples. Accelerometer, seismometer. Vibration of two-degree of freedoms system: Influence coefficient method - numerical examples. Vibration absorbers.	
4	Vibration of multi-degree freedom system: Transverse vibration, concept of node, mode, mode shape. Dunkerley's method, critical or whirling speed of shaft. Torsional vibration of shaft: Torsionally Equivalent shaft - two rotor and three rotor systems, free torsional vibration of geared systems - numerical examples.	9

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental principles of dynamics and their application to machinery; analyse the forces acting on the machine components.	K4
CO2	Apply the principles and applications of flywheels, calculation of gyroscopic angular momentum, torque and stability.	K3
CO3	Analyse and predict the vibrational behaviour of the mechanical system.	K4
CO4	Apply the concepts and analysis of damped vibration and the effects of damping on vibrational behaviour.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	2	-	-	-	-	-	2
CO2	3	3	-	-	-	2	-	-	-	-	-	2
CO3	3	3	-	-	-	2	-	-	-	-	-	2
CO4	3	3	-	-	-	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of Machines	S S Rattan	Tata Mc Graw-Hill	4th Ed., 2017
2	Theory of Vibrations with Applications	W T Thomson	Prentice Hall	5th Ed., 2014
3	Theory of Machines- Kinematics and Dynamics	Sadhu Singh	Pearson Edu India	3rd Ed., 2012
4	Theory of Machines	D S Kumar	Laxmi Publications	4th Ed., 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Vibration and noise for Engineers	Kewal Pujara	Dhanpat Rai & CO	4th Ed., 2007
2	Theory of Machines	Thomas Bevan	Oxford University Press	4th Ed., 2018
3	Theory of Machines and Dynamics of Machinery	J S Rao and D.K Sharma	McGraw Hill Education	3rd Ed., 2019
4	Theory of machines	N K Gupta	Tata McGraw-Hill Education	4th Ed., 2019
5	Theory of Machines	R.K Bansal	Laxmi Publications	3rd Ed., 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/104/112104114/
	https://archive.nptel.ac.in/courses/112/107/112107212/
	https://archive.nptel.ac.in/courses/112/103/112103111/

SEMESTER: S8

INDUSTRIAL SAFETY ENGINEERING

Course Code	PEPET865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the concepts of safety, hazard and its issues.
2. To enable students to apply various safety management and safety audit techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Need for safety, safety and productivity, accident, injury, unsafe act, unsafe condition, theories of accident causation. Safety organization: objectives, types, functions, safety policy, safety officer - responsibilities, authority. Safety committee - need, types, advantages. Personal protection: Types of personal protective equipment (PPE), respiratory and non-respiratory equipment, standards related to PPE.	9
2	Hazards and risk: Types of hazards - classification of fire, types of fire extinguishers, fire explosion and toxic gas release, structure of hazard identification and risk assessment. Identification of hazards: Inventory analysis, fire and explosion hazard rating of process plants, Dow's index, Preliminary hazard analysis, Hazard and Operability (HAZOP) analysis- methodology, criticality analysis, corrective action and follow-up.	9
3	Machinery safeguard: Point-of-operation, principle of machine guarding, types of guards and devices. Safety in process: Safety in turning, grinding, rolling, forging, welding and cutting processes.	9

	Material Handling: Classification, safety consideration - manual and mechanical handling. Handling assessments and techniques - lifting, carrying, pulling, pushing, palletizing and stocking. Material handling equipment - operation & maintenance.	
4	Safety audit: Objectives of safety audit, limitations and benefits of safety audit, types of safety audits - industrial/workplace safety audit, fire safety audit, audit methodology, audit checklist and report, significance of performing a safety audit, conducting safety audit, pre-audit activities, on-site activities, post audit activities, safety inspection vs safety audit.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> <i>Clearly defines the real-world quality issue.</i> <i>Examine and identify relevant contextual factors.</i>	5
	<i>Problem Analysis</i> <i>Present a structured realistic solution methodology.</i> <i>Compare and justify the proposed solutions with evidence and logical reasoning</i>	5
Evaluate	<i>Validation of Results</i> <i>Thoroughly evaluate the proposed solutions.</i> <i>Compares trade-offs, advantages, and disadvantages.</i> <i>Considers feasibility, scalability, and practical implications.</i>	5
	<i>Conclusion and Report Writing</i> <i>Summarizes procedure, findings and insights, limitation, and scope for future work.</i> <i>Preparation of Report with all components of project report.</i>	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of safety and personal protective equipment.	K2
CO2	Demonstrate different types of hazards and apply methods of hazard identification.	K3
CO3	Explain the safety issues in machinery, processes and material handling systems.	K2
CO4	Apply safety audit concepts and methodologies.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping odd Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	1	-	-	3	-	-	-	-	-	2
CO2	3	-	2	-	-	3	-	-	-	-	-	2
CO3	3	-	1	-	-	3	-	-	-	-	-	2
CO4	3	-	2	-	-	3	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Safety, Health and Environment management systems.	R K Jain and Sunil S Rao	Khanna Publications	1st Ed., 2000
2	Safety management System and Documentation training Programme handbook.	Paul S V	CBS Publication	1st Ed., 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety management in Industry	Krishnan N V	Jaico Publishing House	1st Ed., 1996
2	Safety management	John V Grimaldi and Rollin H Simonds	Irwin	5th Ed., 1989
3	Industrial safety	Ronald P Blake	Prentice Hall India	3rd Ed., 2000
4	Safety management system	Alan Waring	Chapman & Hall, England	1st Ed., 1996

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105094/

SEMESTER: S8

RENEWABLE ENERGY ENGINEERING

Course Code	OEPET831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive knowledge of various renewable energy resources and their technological applications.
2. To familiarise students with implementation and environmental impacts of renewable energy technologies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Global energy scenario: Energy demand and consumption, conventional and renewable energy sources. Solar radiation and solar thermal systems: Solar constant, spectral distribution, measurement of solar radiation. Solar thermal systems - flat plate collectors, concentrating collectors, solar water heaters, solar dryers, solar cooling systems. Solar electrical systems: Photovoltaic (PV) technology - PV cells and materials, PV modules and arrays, PV system design and applications, maximum power point tracking, solar cell characteristics, stand-alone and grid-connected PV systems.	9
2	Wind energy fundamentals: Wind characteristics, wind power equation, power extraction from wind. Wind turbines - types of wind turbines, horizontal and vertical axis wind turbines, components and operation, wind farm layout and design. Wind power generation - stand-alone and grid-connected wind systems, hybrid systems. Bio-energy: Types of biomass, availability and potential, biomass conversion technologies - thermo-chemical processes, biochemical processes, applications of biomass in power generation, heating, and transportation.	9

3	Geothermal energy: Geothermal resources and types of geothermal power plants - dry steam, flash steam, binary cycle. Ocean energy: Ocean thermal energy conversion, tidal energy, types of tidal power plants, wave energy and wave energy converters. Hydropower: Small, mini and micro hydropower plants, components and environmental impacts. Fuel cells: Fundamentals, working principles of fuel cells, types of fuel cells.	9
4	Environmental impact of renewable energy: Impact on ecosystems, carbon footprint reduction, carbon credits. Economics of renewable energy systems: Cost analysis, payback period, life-cycle analysis and costing. Renewable energy policies and regulations: National and international energy policies, renewable energy targets, government schemes and incentives - India's national solar mission, renewable energy purchase obligations.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of renewable energy, solar radiation, thermal systems and electrical systems.	K2
CO2	Demonstrate wind energy fundamentals and bio energy.	K2
CO3	Explain geothermal, ocean and hydropower energy systems as well as fuel cell technologies.	K2
CO4	Demonstrate environmental and economic impact of renewable energy systems; explain renewable energy policies and regulations.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	1	-	-	2	2	-	-	-	-	2
CO2	3	-	2	-	-	2	2	-	-	-	-	2
CO3	3	-	2	-	-	2	2	-	-	-	-	2
CO4	3	-	2	-	-	2	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Renewable Energy Sources and Emerging Technologies	D P Kothari, K C Singal and Rakesh Ranjan	PHI Learning Pvt Ltd	3rd Ed., 2021
2	Renewable Energy: Power for a Sustainable Future	Stephen Peake	OUP Oxford	4th Ed., 2017
3	Energy Storage for Renewable Energy Systems	Behnam Zakeri and Sanna Syri	Elsevier	1st Ed., 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Renewable Energy Processes	Akwas Boateng	Academic Press Inc	1st Ed., 2017
2	Energy Management	Murphy W. R.	Elsevier/bsp Books Pvt. Ltd.	1st Ed., 2003

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/103/103/103103206/

SEMESTER: S8

OCCUPATIONAL SAFETY AND HEALTH

Course Code	OEPET832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the aspects of occupational safety and health in industries.
2. To equip students to apply various techniques available for ensuring safety in industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Safety, quality and productivity, safety organizations, safety committees and structure, role of management in industrial safety, occupational safety and health act, environmental management systems and ISO 14001. Industrial hygiene: History, housekeeping, OSHA and industrial hygiene, safety and health program management, education and training for industrial safety and hygiene.	9
2	Fire hazards and fire extinguishers: Fire chemistry, causes of fire, classes of fire, principles of fire extinction, water sprinkler and fire hydrant, explosion protection system, alarm and detection system, suppression system, carbon dioxide system, foam system, halon system, portable extinguisher.	9
3	Workplace stressors and associated hazards: Chemical stressor, physical stressor, biological stressor, ergonomic stressor, noise, vibration and their control, radiation safety program, thermal stressors and their control, ventilation investigation and analysis. Industrial toxicology: Material safety data sheet, hazard communication standard, airborne contaminants, air monitoring and sampling.	9

4	Hazard identification methods: Safety inspection, job safety analysis, risk analysis methods - map method and tabular method, fault tree analysis, HAZOP analysis. Hazard control: Hierarchy of control - elimination, substitution, engineering control, administrative control, personal protection equipment (PPE), training, maintenance and storage of PPE, accident precautions, economic aspects of accidents, accident reporting.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain basic aspects of occupational safety and industrial hygiene.	K2
CO2	Demonstrate fire hazards and extinguishing methods.	K2
CO3	Explain workplace stressors and associated hazards.	K2
CO4	Apply hazard identification methods and hazard control.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	-	3	-	-	-	-	-	2
CO2	3	-	2	-	-	3	-	-	-	-	-	2
CO3	3	-	2	-	-	3	-	-	-	-	-	2
CO4	3	-	3	-	-	3	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Safety	Blake R P	Prentice Hall	3rd Ed., 2006
2	Industrial Safety Management	Deshmukh L M	Tata McGraw Hill	3rd Ed., 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Safety, Health and Environmental Management Systems	Jain R K and Sunil Rao	Khanna Delhi	1st Ed., 2006
2	Chemical Hazards and Safety	Shrikant Dawande	Khanna Publishers	2nd Ed., 2012
3	Chemical Process Safety: Fundamentals with applications	Daniel A Crowl and Joseph F Louvar	Prentice Hall	3rd Ed., 2011
4	Safety management	John V Grimaldi and Rollin H Simonds	Irwin	5th Ed., 1989
5	Industrial Hygiene Simplified: A Guide to Anticipation, Recognition, Evaluation, and Control of Workplace Hazards	Frank R Spellman	Bernan Press	2nd Ed., 2017
6	Industrial Safety	Blake R B	Prentice Hall	3rd Ed., 2000
7	Techniques of Safety Management: A system Approach	Dan Petersen	Amer Society of Safety Engineers	4th Ed., 2003

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105094/

SEMESTER: S8

DIGITAL MARKETING

Course Code	OEPET833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the basics of digital marketing principles, strategies and tools.
2. To enable students to apply the practical skills in digital marketing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Digital marketing foundations: Introduction to digital marketing, traditional marketing vs. digital marketing, digital marketing process, website planning and development, domain and web hosting, building websites using CMS WordPress, wordPress plug-ins. Search engine optimization (SEO): Introduction, keyword planner tools.	9
2	Digital marketing optimization: On-page SEO techniques, content optimization, Yoast SEO plug-in, off-page SEO techniques. Email marketing: Introduction, designing email marketing campaigns using mail chimp, building email lists and signup forms, email marketing strategy and monitoring and email automation.	9
3	Digital advertising and analytics: Pay per click advertising, introduction, google AdWords, bidding strategies, designing and monitoring search campaigns, display campaigns, video campaigns, universal app campaigns, Google Analytics - introduction, interface and setup.	9
4	Social media and integrated digital marketing: Social media marketing introduction, facebook marketing, twitter marketing, LinkedIn marketing, developing digital marketing strategy, integrating digital marketing channels, measuring digital marketing success, digital marketing case studies.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the principles and concepts of digital marketing.	K2
CO2	Apply SEO techniques for content optimization and explain the concepts of email marketing.	K3
CO3	Make use of digital advertising and Google Analytics for developing marketing strategies.	K3
CO4	Demonstrate social media and integrated digital marketing.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	-	-	-	-	-	-	-	2
CO2	3	-	2	-	2	-	-	-	-	-	-	2
CO3	3	-	2	-	3	-	-	-	-	-	-	2
CO4	2	-	-	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Marketing	Seema Gupta	McGraw Hill	3rd Ed., 2022
2	Digital Marketing Strategy: An Integrated Approach to Online Marketing	Simon Kingsnorth	Kogan page publications	2nd Ed., 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Marketing: Strategy, Implementation, and Practice	Dave Chaffey	Pearson Education	1st Ed., 2022
2	Digital Marketing	Kamlesh Kudacya	McGraw Hill Education	1st Ed., 2022
3	Digital Marketing: A Practical Approach	Anita Goyal	Pearson Education	1st Ed., 2020
4	Inbound Marketing	Brian Halligan, Dharmesh Shah	Wiley	2nd Ed., 2014
5	Content Marketing: A Practical Approach	Pamela Muldoon	Routledge	1st Ed., 2020
6	Email Marketing By the Numbers	Chris Baggott	Wiley	1st Ed., 2017
7	Google Analytics Breakthrough	Feras Alhlou	Wiley	1st Ed., 2020
8	Social Media Marketing: An Hour a Day	Dave Evans	Wiley	1st Ed., 2018
9	Digital Marketing Analytics	Chuck Hemann	Routledge	1st Ed., 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://onlinecourses.swayam2.ac.in/imb25_mg31/
	https://onlinecourses.swayam2.ac.in/cec19_mg23/